A Conditional Reconstruction Program for Relativity and Standard Model Structure from Observer-Overlap Consistency

Observers Are All You Need (Compact)

B. Müller Peter Nguyen

March 28, 2026

Paper release: `r1352` **Released:** March 28, 2026

Abstract

We present a compact, falsifiable Phase-I statement of Observer-Patch Holography (OPH) as a conditional reconstruction program whose claimed recovered core is limited to relativity and Standard Model structure. From five axioms, the stated scaling-limit/categorical regularity package, and two external continuous inputs used only in the current quantitative implementation, the compact note derives a schedule-independent normal form for overlap consistency, a conditional Lorentz branch on the screen, a conditional Jacobson-type Einstein branch from fixed-cap generalized-entropy stationarity together with the null modular bridge and the scaling-limit null-stress premise, and conditional compact gauge reconstruction in the bosonic internal-gauge branch from a refinement-stable symmetric edge-sector colimit. After Axiom 3.5 (Minimal Admissible Realization, MAR) is imposed on the connected positive-dimensional Lie admissible class with one connected abelian charge factor, the realized low-energy gauge group on that realized branch is

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

together with the exact rational hypercharge lattice for the realized one-generation Standard Model chiral matter package with one Higgs doublet. The realized counting chain is sequential rather than joint: on the realized MAR-admissible branch, the asymptotic-freedom/CP-capability window and minimal admissibility yield $N_g = 3$, and then Witten's SU(2) anomaly parity constraint together with the same realized-branch selector yield $N_c = 3$. The two external continuous inputs of the present implementation are the pixel area

$$P \equiv a_{\mathrm{cell}}/\ell_P^2 = 1.63094$$

and the total screen capacity

$$N_{\mathrm{scr}} \equiv \log\dim\mathcal{H}_{\mathrm{tot}} \sim 10^{122}.$$

In the current implementation, P feeds a supplement-backed gauge-coupling calibration sector and N_{scr} feeds the capacity branch for Λ; these non-core numerical branches are reported only as implementation-level checks, not as the recovered core itself. Charged-lepton/Koide, texture, capacity-level neutrino estimates, dark-sector, baryogenesis, black-hole-spectroscopy, proton-spin, proton-lifetime estimates beyond the structural exclusion of gauge-mediated decay, and string/worldsheet discussions are explicitly deferred continuations requiring additional ansätze beyond the recovered core and are not claimed as outputs of overlap consistency plus the stated

scaling-limit and categorical premises. The purpose of this compact note is therefore to state a clean Phase-I claim set: relativity and Standard Model structure are the recovered core, while all downstream branches are either input-dependent or supplement-backed secondary sectors, or post-Phase-I phenomenology.

1 Introduction

The standard model of particle physics and general relativity are successful effective descriptions. They leave open a familiar list of questions: why the Lorentz group appears, why the low-energy spacetime description follows a Jacobson-type Einstein branch, why the gauge structure is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

why the observed three generations and three colors are realized, why hypercharges take the observed rational values, what fixes the electroweak and flavor scales, and what the present framework can and cannot say about the cosmological constant and possible dark-sector continuations.

This paper presents a single framework that addresses this list as a conditional reconstruction program. The framework is Observer-Patch Holography (OPH). Its starting point is an observer-centric organization of physical data: each observer has access only to a patch algebra, neighboring observers must agree on overlaps, and global physical law is recovered from the resulting consistency conditions. The same formal structure that produces schedule-independent overlap consistency then feeds a Lorentzian modular-phase classification, a Jacobson-type Einstein branch conditional on fixed-cap stationarity and the null-stress premise, compact gauge reconstruction in a bosonic internal-gauge sector, charge quantization, and the low-energy architecture selected by the admissibility conditions used in the gauge branch.

Two decisions shape the present version.

1. The compact note presents the main theorem statements in one place, but some quantitative conventions and implementation details live in repository supplements.[1] For length, several theorem-level arguments below are recorded only as proof sketches; the synchronized main paper contains the full derivations.

2. The paper separates exact structural theorems, conditional scaling-limit branches, calibration-sector outputs, and phenomenological continuations. This keeps strong claims visible and weak claims identifiable.

The framework is intended as a candidate fundamental theory in the precise sense relevant for physics: general relativity and the Standard Model appear as effective sectors derived from a smaller axiom set plus an explicit refinement/scaling-limit package. The paper does not use simulation language, observer continuation stories, or other non-falsifiable additions.

Prioritized achievements

The results emphasized in this paper are ordered below by a combination of impact and defensibility.

1. A conditional Jacobson-type Einstein branch is recovered from observer-overlap modular geometry, an explicit fixed-cap generalized-entropy stationarity premise, the geometric-branch

[1] Broader interpretive questions are deferred here so the compact note can keep its claim-tier boundary explicit.

selection burden, and an explicit null-modular / null-stress scaling-limit package; in the null modular bridge, quasi-local propagation, endpoint control, half-sided inclusion, and the explicit positive null-translation generator are now supplied internally through the derived Borchers stage, while the density upgrade and null-stress identification remain explicit downstream burdens.

2. The exact Standard Model gauge structure

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

is recovered in the bosonic compact-gauge branch once the MAR admissibility package is imposed.

3. On the realized one-generation chiral matter plus one-Higgs package, anomaly constraints and Yukawa invariance fix the hypercharge lattice, and on the realized MAR-admissible branch the counting chain yields $N_g = 3$ and then, using Witten's SU(2) anomaly parity constraint, $N_c = 3$.

4. The current quantitative implementation uses two external continuous configuration inputs, P and N_{scr}.

5. Downstream matter-sector continuations, including charged-lepton/Koide structure, remain visible but explicitly outside the compact paper's recovered-core theorem package.

6. A separate input-dependent capacity corollary ties the cosmological-constant branch to global screen capacity rather than local vacuum energy.

7. Exclusion of gauge-mediated proton decay follows from product-group structure; any stronger proton-lifetime claim remains UV/EFT/QCD-dependent.

8. Dark-sector, baryogenesis, string, and spectroscopy branches are kept visible, but only as program-level continuations or, where additional ansätze are involved, conjectural phenomenology.

9. Overlap repair admits a unique schedule-independent normal form on the stated finite patch-net hypotheses.

2 Results at a Glance

This section is the authoritative status ledger for the compact note and the sync rule for mirrored summary surfaces. The three-phase boundary used throughout is

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10, \qquad \text{Phase III} = D12.$$

Phase I is the recovered-core claim set. Phase II contains the separate input-dependent capacity corollary and the current supplement-backed gauge-calibration sector. Phase III collects deferred continuations. The canonical five-axiom basis, the two external continuous inputs, and the technical-premise labels used below are fixed in Section 3. Each row names a node in the reconstruction program, records its immediate OPH-side parents, separates imported standard mathematics from extra non-axiom physical premises or external inputs, and states the resulting claim tier.

Table 1: Authoritative dependency checklist for the compact OPH program.

Node	Output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Claim tier
D1	Unique schedule-independent normal form (Theorem 4.3)	overlap-consistency problem on a finite patch net	Newman's lemma for terminating locally confluent rewrite systems	finite state spaces, termination, and local confluence	Phase I structural theorem
D2	Collar Markov/entropy split and G-dictionary setup (Theorem 5.2, Propositions 5.6, 5.7, 5.10, Definition 5.11)	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact block-factorized identities require exact Markovity or an explicitly stated idealized limit; the $A/(4G)$ step is the coarse-grained dictionary of Definition 5.11	Phase I structural lemma/proposition layer
D3	Lorentz branch $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ (Theorem 6.1, Corollary 6.3)	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; conformal-group classification	T4 together with T8, with the standard cap-dilation identification carried directly by the BW geometric branch in T4	Phase I conditional scaling-limit theorem
D4	Null modular bridge to T_{kk} (Propositions 6.4, 6.7; Corollaries 6.5, 6.9; Theorem 6.8, Lemma 6.10; Theorem 6.11)	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock half-sided modular inclusion; distributional differentiation on half-lines	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Proposition 6.4 and Corollary 6.5, plus T8 together with the density-upgrade hypotheses of Theorem 6.8 and the relativistic null-stress identification premise; the propagation controls denoted T2–T3 are supplied internally by Axiom 3.3 on the local finite-constraint branch	Phase I conditional bridge theorem
D5	Jacobson-type Einstein branch: rest-frame relation plus conditional tensor upgrade (Lemma 6.16, Theorem 6.17, Corollary 6.18)	D3+D4 and Axioms 3.1–3.4	Jacobson small-ball first law; null-to-tensor reconstruction modulo a metric term	T9 fixed-cap stationarity; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; D4 carried remainder negligible at order ℓ^4; all local directions/reference states for the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	Capacity branch for Λ and the separate order-of-magnitude neutrino estimate (Proposition 6.20, Corollary 6.21)	D5 leaves a $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent corollary / estimate
D7	Compact gauge reconstruction (Theorem 7.1)	Axioms 3.1–3.4	Doplicher–Roberts / Tannaka reconstruction	T7 and T10–T11 together with the directed-colimit/fiber clause of T12; the refinement-stable qualifier on that colimit is supplied internally by Axiom 3.3	Phase I conditional structural theorem

Table 1: Authoritative dependency checklist for the compact OPH program.

Node	Output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Claim tier
D8	Product gauge structure up to finite quotient (Lemmas 7.3–7.8, Theorem 7.9)	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	connected positive-dimensional Lie admissible class; one connected abelian factor; faithful action on the minimal coupled carrier; vanishing obstruction $[z] = 0$	Phase I realized-branch theorem
D9	Realized Standard Model quotient, hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and product-group corollaries (Theorem 7.10, Corollaries 7.11, 7.12, Proposition 7.13)	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package; asymptotic-freedom and MAR admissibility premises	Phase I realized-branch theorem/corollary chain
D10	Gauge-coupling numerical closure of the current supplement-backed implementation	D9 + external input P	RG evolution and matching conventions supplied in the supplement	pixel constraint; edge heat-kernel closure; supplement-backed beta-function and threshold conventions	Phase II calibration sector
D12	Charged-lepton/Koide, texture, dark-sector, baryogenesis, black-hole spectroscopy, proton-spin, proton-lifetime estimates beyond the gauge-channel exclusion, conditional string/worldsheet reorganizations, conjectural critical-superstring extensions, and other continuations	various subsets of D6, D9, and D10	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, discrete texture choices, dark-sector response assumptions, discrete-horizon assumptions, or extra worldsheet/CFT premises	Phase III phenomenological continuation / program-level branch

Accordingly, the recovered-core structural program of the compact note is D1–D5 together with D7–D9. D6 is the separate Phase-II input-dependent corollary; D10 is the current Phase-II supplement-backed calibration sector; and D12 is the Phase-III continuation layer. The modular-anomaly dark-sector branch sits in D12: neither MOND/RAR-style response laws nor galaxy/cluster phenomenology are part of the compact note's theorem-level claim set. The only external continuous inputs are N_{scr} in D6 and P in D10, together with descendants built from those branches.

3 Five Axioms, Two External Continuous Inputs in the Current Quantitative Implementation, and Technical Premises

This section is the authoritative premise ledger and dependency map for the compact note. The OPH basis used below is exactly the five core axioms, the two external continuous inputs of the current quantitative implementation, and the theorem-local technical premises collected afterward. Older A1–A4 / B / MX / LR / R0–R1 packaging is retained elsewhere only as shorthand for pieces of this canonical package, not as a competing axiom basis. In the present version, the local finite-constraint MaxEnt/refinement branch already internalizes the quasi-local propagation and refinement-stability control that older packaging sometimes carried as separate regularity language, Theorem 6.1 fixes the cap generator and its 2π normalization on the OPH Bisognano–Wichmann geometric branch without adding any separate cap-isotropy selector, and Corollary 6.9 derives the null half-sided modular pair inside the synchronized bridge. What remains genuinely theorem-external is the scaling-limit scope clause, the geometric-branch selection burden, the density-upgrade burden, fixed-cap stationarity, the bosonic/fiber-functor fork, the relativistic null-stress identification premise, and the vanishing obstruction premise where invoked.

The formulation uses two external continuous configuration inputs in the current quantitative implementation:

$$P \equiv a_{\mathrm{cell}}/\ell_P^2 = 1.63094, \tag{1}$$

$$N_{\mathrm{scr}} \equiv \log \dim \mathcal{H}_{\mathrm{tot}} \sim 10^{122}. \tag{2}$$

In the current implementation, P is calibrated from the gauge sector and N_{scr} is inferred from the observed cosmological constant. The first sets the particle-physics scale. The second enters the cosmological-capacity branch and a separate order-of-magnitude neutrino estimate.

Axiom 3.1 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 3.2 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 3.3 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale ℓ_{UV}, the realized branch is selected by maximizing entropy subject to a finite family of gauge-invariant local constraints*

$$\mathcal{C}_{\ell_{\mathrm{UV}}} = \{O_a(x)\}_{a=1}^{N_{\mathrm{con}}},$$

where each $O_a(x)$ is supported in a ball of radius $O(\ell_{\mathrm{UV}})$ and the label set a is independent of the number of regulator cells. Under refinement, the same finite constraint family is preserved, so the realized states at different cutoffs belong to one common finite-dimensional MaxEnt family rather than unrelated maximizers. The realized low-energy branch is the refinement-stable branch of that family; therefore symmetry-allowed relevant operators are not held at zero by unexplained fine tuning, but only by symmetry or because they lie in the explicitly retained constraint family.

Derived local-Gibbs branch. On the finite regulator realization of Axiom 3.1, the standard finite-dimensional Lagrange-multiplier argument applied to Axiom 3.3 gives

$$\omega_{\ell_{\mathrm{UV}}}(\lambda) = Z_{\ell_{\mathrm{UV}}}(\lambda)^{-1} \exp\bigl(-K_{\ell_{\mathrm{UV}}}(\lambda)\bigr),$$

with

$$K_{\ell_{\mathrm{UV}}}(\lambda) = \sum_{x} \sum_{a=1}^{N_{\mathrm{con}}} \lambda_a O_a(x) + \sum_{b} \mu_b Q_b,$$

where the Q_b are the finitely many optional global conserved-charge constraints. Thus the logarithm of the selected state is itself a quasi-local UV generator built from the same bounded-support densities that define the branch.

Derived quasi-local propagation and interval control. Let $\tau_t^{\ell_{\mathrm{UV}}}$ denote the automorphism group generated by $K_{\ell_{\mathrm{UV}}}(\lambda)$, or more generally by any branch generator lying in the norm-closed algebra generated by the same bounded-support local densities. Standard Lieb–Robinson estimates for finite-range interactions on the finite regulator net then give constants $C, \xi, v_{\ell_{\mathrm{UV}}} < \infty$ such that for local observables A_X, B_Y,

$$\bigl\|[\tau_t^{\ell_{\mathrm{UV}}}(A_X), B_Y]\bigr\| \le C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\ell_{\mathrm{UV}}}|t|)/\xi}.$$

So finite-velocity quasi-local propagation is branch-internal: it is the local finite-constraint MaxEnt branch written in propagation form, not an extra selector. The same locality statement gives the bounded-interval endpoint control later used in the null modular chain: after removal of the central endpoint term, changing an interval I to I' alters only an $O(|I' \Delta I|)$ collar of local terms. Hence on any fixed local-energy-bounded domain one has

$$\bigl|\langle \psi, (K[I'] - K[I])\phi \rangle\bigr| \le C_{\psi,\phi,I_{\max}} \, |I' \Delta I|,$$

which is precisely the endpoint-Lipschitz / finite-variation control later denoted by **T2**–**T3**. The compact note does *not* separately derive a second independent microscopic generator beyond this quasi-local branch generator; if a UV completion carries one, the only later requirement is that it lie in the same bounded-support algebraic closure so that the same propagation constants control the selected branch. So the fixed-cutoff UV branch is already interacting in the ordinary finite-range sense. This local-Gibbs/Lieb–Robinson package should not be read as a proof that the refinement limit lands in a trivial gauge phase or that OPH has selected a unique microscopic realization; its theorem-level role is collar-local mixing and the carried propagation/error control used by the later scaling-limit arguments.

Internal refinement notion. Because Axiom 3.3 preserves the same finite local constraint family under coarse-graining, the regulator-scale states are parameterized by one common finite-dimensional multiplier space λ. The refinement-stable OPH branch is therefore the fixed point or stable manifold selected inside that multiplier space, not an imported theorem-external regularity package. Any transportable edge sector that persists along this branch defines a canonical directed system under refinement. Accordingly, whenever later sections speak of a "refinement-stable directed colimit" of edge sectors, the refinement-stable qualifier means persistence along the Axiom 3.3 branch itself; only the bosonic/symmetric/fiber-functor clauses used in gauge reconstruction remain additional categorical premises.

Axiom 3.4 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \langle L_C \rangle,$$

*where L_C is a positive edge-center entropy functional. In the semiclassical scaling branch, its leading coarse-grained contribution is identified with $A(\partial C)/(4G)$. The functional obeys quantum focusing on null generators, and collar tripartitions have small conditional mutual information with controlled recovery maps [**?**, **?**, **?**, **?**].*

Axiom 3.5 (Minimal Admissible Realization)**.** *Among realized sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are loop-coherent, anomaly-free, refinement-stable with light chiral matter, single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsically CP-capable, and weak-sector UV-completable, the realized package is the lexicographically minimal one under*

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\, N_{\mathrm{nonab}},\, N_c,\, N_g).$$

$\mathfrak{S}$ is the sector package on which MAR acts; it is not the bare tensor category alone. MAR is therefore an explicit structural-economy axiom on already-admissible realized low-energy branches, not a theorem derived from the preceding axioms. χ_{cpl} is the coupled carrier dimension: the smallest unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is stronger than the abstract minimal faithful representation dimension.

Assumption 3.6 (Technical Premises)**.** *The continuum reductions still use several historical shorthand labels, but not all of them are extra premises, and the retained numbering is intentionally nonconsecutive for compatibility with earlier drafts and review surfaces. The labels **T2**–**T3** now refer only to the branch-internal quasi-local propagation and bounded-interval endpoint control already derived from Axiom 3.3; likewise, whenever later sections speak of a "refinement-stable" sector colimit, that qualifier means persistence along the same Axiom 3.3 branch rather than an additional categorical regularity axiom. The old label **T6** is retained only as legacy shorthand for the derived half-sided modular-inclusion conclusion of Corollary 6.9; it is no longer an extra premise. The genuinely theorem-external items retained below are therefore eight in number:*

- ***T1*** *type-I local algebras at the regulator scale and boundary gauge invariance on collars;*
- ***T4*** *the branch condition that the refinement-stable cap net lies in the Bisognano–Wichmann geometric modular class for caps and their conformal images: for each round cap $C \subset S^2$, the scaling-limit modular automorphism group of $\mathcal{A}(C)$ is implemented by the standard cap-preserving conformal one-parameter subgroup $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$, with geometric parameter s normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$; no separate cap-isotropy / $\mathrm{SO}(2)$-equivariance selector is imposed;*
- ***T7*** *vanishing gluing obstruction $[z] = 0$ whenever global sector transportability is invoked;*
- ***T8*** *all Lorentz/null-modular/Einstein statements are claims about a controlled refinement/scaling limit of the regulator net, not literal exactness at fixed finite cutoff;*
- ***T9*** *fixed-cap generalized-entropy stationarity for the admissible first-variation class used in the Jacobson branch;*

T10 *symmetric braiding in the* $3+1D$ *EFT branch;*

T11 *a bosonic Tannakian fiber-functor premise, or else an explicit super-Tannakian fork;*

T12 *whenever compact gauge reconstruction is invoked, the sector category means the directed colimit of transportable edge sectors with objectwise finite-dimensional fibers.*

Remark 3.7 (Legacy shorthand)**.** *Historical labels are read as follows in the compact note and in mirrored surfaces: A1/A2 mean Screen Net and Overlap Consistency; B together with refinement stability means Axiom 3.3; A3 means Axiom 3.4; A4 together with the old MX / collar terminology means the recoverability and collar-control ingredients explicitly cited from Axiom 3.4; LR or the later* ***T2–T3*** *shorthand now mean the derived quasi-local propagation and endpoint control of the local finite-constraint MaxEnt branch, not an extra premise; and R0/R1 and* $[z] = 0$ *remain theorem-local technical premises.*

Remark 3.8. *The five axioms fix the OPH core, and in the present packaging they already internalize the quasi-local propagation and refinement-stable notion of sector persistence used by the scaling-limit and gauge branches. On the gravity side this means the null bridge no longer imports a separate propagation, endpoint-control, or half-sided-inclusion premise: c.12 closes the fixed-cutoff strip argument through exact-or-controlled strip additivity, the weak tail generator, and the derived half-sided modular pair whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain. The cap-modular geometric action and the* 2π *KMS normalization are recovered in Theorem 6.1 once the OPH Bisognano–Wichmann geometric branch of* ***T4*** *is imposed; what remains genuinely theorem-external is the retained eight-item technical package: the obstruction clause* ***T7*** *when transportability is invoked, the scaling-limit scope clause* ***T8****, fixed-cap stationarity* ***T9****, and the bosonic/symmetric/fiber-functor part of* ***T10–T12****, together with* ***T1*** *and* ***T4*** *where the fixed-cutoff regulator presentation and the geometric cap-modular branch hypothesis are used. In particular, the Lorentz/null-modular/Einstein and compact-gauge statements below are still scaling-limit claims rather than fixed-cutoff matrix-algebra identities.*

Remark 3.9 (Log convention)**.** *Unless an explicit base is written, logarithms and entropies in this compact note use natural logs (nats). Base-2 logarithms are written as* $\log_2$ *and quoted in bits.*

Authoritative dependency DAG

The table below is the authoritative dependency map for the compact note. The "Immediate OPH ingredients" column records only parents internal to the OPH program. Imported standard mathematics and extra non-axiom physical premises or external inputs are stated separately rather than hidden inside the word "derived."

Node	Compact theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D1	Theorem 4.3: unique normal form and schedule independence	overlap-consistency problem on a finite patch net	Newman's lemma for terminating locally confluent rewrite systems	finite state spaces, termination, and local confluence	structural theorem

Node	Compact theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D2	Theorem 5.2, Propositions 5.6, 5.7, 5.10, Definition 5.11: collar Markov/entropy layer	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact Markov identities require exact Markovity or an explicitly stated idealized recoverability limit; the $A/(4G)$ identification is the coarse-grained dictionary of Definition 5.11	structural lemma/proposition layer
D3	Theorem 6.1, Corollary 6.3: geometric modular flow and Lorentz branch	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$	T4 together with T8, with the standard cap-dilation identification carried directly by the BW geometric branch in T4	conditional scaling-limit theorem
D4	Propositions 6.4, 6.7; Corollaries 6.5, 6.9; Theorem 6.8, Lemma 6.10; Theorem 6.11: null modular bridge to T_{kk}	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock half-sided modular inclusion; distributional differentiation on half-lines	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Proposition 6.4 and Corollary 6.5, plus T8 together with the density-upgrade hypotheses of Theorem 6.8 and the relativistic null-stress identification premise; the propagation controls denoted T2–T3 are supplied internally by Axiom 3.3 on the local finite-constraint branch	conditional bridge theorem
D5	Lemma 6.16, Theorem 6.17, Corollary 6.18: Jacobson-type Einstein branch via a rest-frame relation plus conditional tensor upgrade	D3+D4 and Axioms 3.1–3.4	Jacobson small-ball first law; Lemma 6.12 for the null-to-tensor upgrade modulo a metric term	T9 fixed-cap stationarity; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; D4 carried remainder negligible at order ℓ^4; all local directions/reference states for the tensor upgrade	conditional scaling-limit theo-rem/corollary
D6	Proposition 6.20, Corollary 6.21: capacity branch for Λ, plus the separate neutrino estimate	D5, which already leaves the $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	input-dependent corollary / estimate
D7	Theorem 7.1: compact gauge reconstruction	Axioms 3.1–3.4	Doplicher–Roberts / Tannaka reconstruction	T7 and T10–T11 together with the directed-colimit/fiber clause of T12; the refinement-stable qualifier on that colimit is supplied internally by Axiom 3.3	conditional structural theorem
D8	Lemmas 7.3–7.8, Theorem 7.9: product gauge structure up to finite quotient	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	connected positive-dimensional Lie admissible class; one connected abelian factor; faithful action on the minimal coupled carrier; vanishing obstruction $[z] = 0$	realized-branch theorem

Node	Compact theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D9	Theorem 7.10, Corollaries 7.11, 7.12, Proposition 7.13: realized Standard Model quotient, hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and product-group corollaries	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package; asymptotic-freedom and MAR admissibility premises	realized-branch theorem/corollary chain
D10	gauge-coupling numerical closure of Section 7	D9 + external input P	RG evolution and matching conventions supplied in the supplement	pixel constraint; edge heat-kernel closure; supplement-backed beta-function and threshold conventions	calibration sector
D12	Sections 8–10 program continuations, including charged-lepton/Koide structure, proton-spin and non-gauge proton-lifetime estimates together with the string/worldsheet branch	various subsets of D6, D9, and D10	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, texture choices, dark-sector response assumptions, discrete-horizon assumptions, or extra worldsheet/CFT premises	phenomenological continuation / program-level branch

No theorem in the compact note is meant to hide standard mathematical imports or extra physical premises inside the phrase "from the axioms." The authoritative unpacking of every such claim is the dependency DAG above. In particular, quasi-local propagation and the refinement-stable sector-persistence notion are now internal to the local finite-constraint MaxEnt branch, the cap generator and its 2π normalization are packaged inside Theorem 6.1 on the OPH geometric branch, the half-sided modular pair is derived inside Corollary 6.9, and the remaining explicit gravity-side burdens are the downstream density-upgrade, fixed-cap stationarity, and null-stress-identification hypotheses. The only rows that use the two external continuous inputs are D6, D10, the separate order-of-magnitude neutrino estimate attached to D6, and downstream branches built from them.

Theorem 3.10 (Summary theorem for the recovered relativity-plus-Standard-Model core)**.** *This theorem packages the recovered-core nodes D1–D5 and D7–D9 of the authoritative dependency DAG above. Node D6 is a separate input-dependent corollary, while D10 and D12 remain outside this theorem's scope. Assume Axioms 3.1–3.4, Axiom 3.5, Assumption 3.6, and the hypotheses of Theorems 4.3, 6.1, 6.11, 6.17, 7.1, 7.9, and 7.10, together with Corollaries 7.11, 7.12, and Proposition 7.13. Then, in the controlled refinement/scaling regime singled out by Assumption 3.6:*

(i) overlap repair admits a unique schedule-independent normal form;

(ii) cap modular flow is geometric in the assumed modular phase and yields the connected Lorentz group

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1);$$

(iii) admissible fixed-cap stationarity together with the null modular bridge and null-stress premise yield the Jacobson-type rest-frame relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame; if that rest-frame relation holds for all local directions and reference states in the scaling regime, then the null-to-tensor upgrade gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

(iv) the refinement-stable transportable edge-sector category reconstructs a compact gauge group, and the realized connected gauge structure has the form

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ.

After the hypercharge lattice, generation-count, color-count, and trivial-action quotient steps, the finite quotient is fixed to $\Gamma = \mathbb{Z}_6$*, so the realized gauge structure on the realized MAR-admissible branch is*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

with the exact Standard Model hypercharge lattice on the realized matter package. Here the realized-branch counting chain is sequential: the asymptotic-freedom/CP-capability window and minimal admissibility first yield $N_g = 3$*, and then Witten's* $\mathrm{SU}(2)$ *anomaly parity constraint together with the same realized-branch selector yield* $N_c = 3$.

Proof. Items (i)–(iii) are collected from Theorem 4.3, Corollary 6.3, Theorem 6.17, and Corollary 6.18. The compact gauge reconstruction is Theorem 7.1; the product gauge structure up to finite quotient is Theorem 7.9; and the final identification of the exact Standard Model quotient is Theorem 7.10 together with Corollaries 7.11, 7.12, and Proposition 7.13. □

Theorem 3.10 therefore packages only the recovered relativity-plus-Standard-Model core: D1–D5 together with D7–D9. The D6 capacity relation is a separate input-dependent corollary; the calibration sector and phenomenological continuations remain outside its scope.

Usually Postulated Elsewhere, Recovered or Continued Here with Explicit Status

Structure	Common treatment	OPH treatment
Lorentz kinematics	background symmetry or starting axiom	conditional scaling-limit branch from geometric modular flow on screen caps
Einstein dynamics	fundamental field equation	conditional first-variation relation with tensor upgrade from generalized entropy, null modular data, and entanglement equilibrium
Gauge group	model input	conditional compact group reconstruction from edge sectors; exact SM quotient selected on the realized MAR-admissible branch
Hypercharge lattice	matter-assignment input	solved from anomaly cancellation and Yukawa invariance on the realized one-generation chiral matter plus one-Higgs package
Color and generation count	empirical input	fixed on the realized MAR-admissible branch by anomaly, CP capability, asymptotic freedom, and minimality
Flavor hierarchy unit	model-dependent small parameter	phenomenological ansatz $\varepsilon = 1/6$ motivated by a uniform $\mathbb{Z}_6$ center-label ensemble
Koide phase	phenomenological fit	continuation ansatz tied to the previously fixed discrete data as $\delta = 2/9$
Cosmological constant	local vacuum-energy puzzle	global screen-capacity term after the input-dependent capacity identification

4 Observer-Overlap Consistency as a Fixed-Point Problem

The discrete overlap problem is the mathematical skeleton of the framework.

Definition 4.1 (Patch Net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ carries a finite local state space S_i. For each edge $e = \{i, j\}$ let I_e be an interface alphabet and let*

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be interface projections. The global state space is

$$\Sigma = \prod_{i \in V} S_i,$$

and the consistency set is

$$C = \{s \in \Sigma : \pi_{i,e}(s_i) = \pi_{j,e}(s_j) \text{ for all } e = \{i, j\} \in E\}.$$

This finiteness is a regulator-level assumption for the discrete patch-net normal-form theorem; it is not the continuum limit statement.

Define the inconsistency potential

$$\Phi(s) = \sum_{e=\{i,j\} \in E} w_e \, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big),$$

with $w_e > 0$ and $d_e(a, b) = 0$ iff $a = b$.

Definition 4.2 (Local Repair Law)**.** *A law λ is a family of local repair maps*

$$T_i^\lambda : \Sigma \to \Sigma$$

changing only patch i or a bounded neighborhood of i.

Theorem 4.3 (Asynchronous Fixed-Point Law)**.** *Assume:*

(i) normal forms of the repair relation are exactly the globally consistent states C;

(ii) the repair relation terminates;

(iii) the repair relation is locally confluent.

Then every $s \in \Sigma$ has a unique normal form

$$\mathrm{nf}_\lambda(s) \in C,$$

and every maximal asynchronous repair sequence terminates at that same state. A sufficient condition for (ii) is that each S_i is finite and every enabled repair strictly decreases Φ.

Proof. Termination is assumed. Local confluence plus termination imply confluence by Newman's lemma. The resulting normal form is unique. By assumption it is a globally consistent state. □

Corollary 4.4 (Schedule Independence)**.** *If observables are evaluated on $\mathrm{nf}_\lambda(s)$, the resulting physical law is independent of update order.*

Theorem 4.5 (Cycle Obstruction / Holonomy Criterion)**.** *Let A be an abelian group. For each oriented edge $e : u \to v$ assign $b_e \in A$ and consider the affine overlap equations*

$$x_v - x_u = b_e.$$

A global solution exists if and only if the signed sum of b_e vanishes on every cycle.

Proof. Summing the edge equations around a cycle gives necessity. For sufficiency, fix a root and define each x_v by summing labels along a path from the root to v. Vanishing cycle sums make this independent of the chosen path. □

Proposition 4.6 (Gauge Quotient)**.** *Suppose a local gauge group $\Gamma = \prod_i \Gamma_i$ acts on Σ while leaving all interface data invariant, and the repair maps satisfy*

$$T_i^\lambda(\gamma \cdot s) = \gamma \cdot T_i^\lambda(s).$$

Then

$$\mathrm{nf}_\lambda(\gamma \cdot s) = \gamma \cdot \mathrm{nf}_\lambda(s).$$

Hence gauge-invariant observables are unique on the quotient.

Proof. Gauge covariance maps repair sequences to repair sequences. Since the normal form is unique by Theorem 4.3, the normal form of $\gamma \cdot s$ must be $\gamma \cdot \mathrm{nf}_\lambda(s)$. □

These three results already encode much of the eventual physical interpretation. Objectivity becomes confluence, gauge symmetry becomes quotient invariance, and stable defects are represented by nontrivial overlap holonomy classes. The physically relevant uniqueness statement is therefore quotient uniqueness: OPH fixes a unique gauge-invariant normal form modulo boundary redundancy and implementation hiding, not a unique microscopic regulator presentation.

5 Collars, Edge Centers, and Generalized Entropy

The fixed-point picture explains why overlap consistency produces gauge quotients, but the gravity and consensus arguments later use a more specific collar fact: after edge-center completion, interior observables become insensitive to compatible exterior substitutions, and modular additivity becomes exact in the Markov normal form. The point of this section is therefore twofold. First, at fixed regulator scale and on the ordinary or central-defect branch, we derive the collar block decomposition from overlap consistency itself. Second, we state precisely when the approximate recoverability premise of Axiom 3.4 is allowed to converge to the exact HJPW normal form used by the later spatial and null collar theorems.

The sharp claim boundary is as follows. Small conditional mutual information always gives a constructive recovered comparison state with $O(\varepsilon^{1/2})$ observable error. It does *not* by itself give a universal one-shot trace-norm bound to an exact Markov state. The exact Markov normal form is recovered either when $I(A : D \mid B) = 0$ holds literally, or for a controlled family on one fixed finite-dimensional collar model, or after pullback to such a model, where $\varepsilon \to 0$. On that fixed collar one gets a collar-dependent modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ measuring distance to the exact Markov set, and this is the error that must be carried into later exact splice or modular-additivity identities.

Proposition 5.1 (Derived regulator gluing datum)**.** *Fix a regulator-scale finite patch cover of a cap neighborhood and let R be any finite union of regulator cells. Assume only the finite patch-net presentation implicit in Definition 4.1 together with overlap consistency on common interfaces. Then:*

1. *each regulator cell i with finite local state space S_i can be Hilbertized as $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{|S_i|}$, so the extended algebra before overlap quotienting is the finite type-I algebra*

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R := \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i;$$

2. *for each connected boundary component $\Sigma \subset \partial R$, after choosing a reference cut presentation $\tilde{\mathcal{H}}_\Sigma$, every overlap-consistent change of local chart along Σ is implemented by a unitary on $\tilde{\mathcal{H}}_\Sigma$, and the compact closure of the subgroup generated by all such recharting unitaries is a compact boundary redundancy group*

$$K_\Sigma \subset U(\tilde{\mathcal{H}}_\Sigma);$$

   before the choice of reference chart, the same data form a compact unitary groupoid of overlap-preserving transitions;

3. *if triple-overlap defects are central, the projective composition law of item 2 lifts to an honest action of a compact central extension $\widehat{K}_\Sigma$; a genuinely noncentral defect is the only obstruction to reducing the transition system to an ordinary compact group action;*

4. *in the invariant-state realization of the quotient, physical observables are the chart-independent endomorphisms, equivalently the fixed-point algebra*

$$\mathcal{A}(R) = \tilde{\mathcal{A}}(R)^{\widehat{K}_{\partial R}}, \qquad \widehat{K}_{\partial R} := \prod_{\Sigma \subset \partial R} \widehat{K}_\Sigma,$$

   with the convention $\widehat{K}_\Sigma = K_\Sigma$ whenever no central extension is needed.

Historical labels $R0$ and $R1$ are shorthand for items 1 and 4 on that branch; they are not extra axioms beyond the finite regulator presentation plus overlap redundancy.

Proof sketch. Item 1 is the Hilbertization of finite sets. For item 2, each overlap-consistent recharting acts on a finite-dimensional matrix algebra, and every *-automorphism of such an algebra is inner. After transporting all local overlap presentations to a chosen reference chart, each recharting is therefore implemented by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group, which yields K_Σ; without fixing the reference chart one has the corresponding compact unitary groupoid. Item 3 is the usual projective-versus-central-extension lift for a central 2-cocycle. Item 4 is the quotient statement: physical observables are precisely the endomorphisms insensitive to recharting of the boundary data, hence the fixed points of the derived boundary action. If the defect is genuinely noncentral, the quotient is still well defined, but the correct bookkeeping object is the crossed-module or higher-gauge data rather than an ordinary compact group. □

Theorem 5.2 (Derived edge-center collar decomposition and exact Markov normal form)**.** *Let A-B-D be a collar tripartition realized at fixed regulator scale on the ordinary or central-defect branch of Proposition 5.1. Write $B = B_L \cup B_R$ with common interface $\Sigma = \partial C$, and let $\widehat{K}_\Sigma$ be the derived compact boundary action attached to that cut, with $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. In the invariant-state realization,*

$$\mathcal{H}_B = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma}.$$

Decompose the left boundary data as

$$\tilde{\mathcal{H}}_{B_L} \cong \bigoplus_\alpha W_\alpha \otimes \mathcal{H}_{b_L^\alpha},$$

with W_α irreducible $\widehat{K}_\Sigma$-modules. Because the right half-collar carries the inverse overlap transport across the same cut, it decomposes contragrediently:

$$\tilde{\mathcal{H}}_{B_R} \cong \bigoplus_\beta W_\beta^* \otimes \mathcal{H}_{b_R^\beta}.$$

Then

$$\mathcal{H}_B \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}, \qquad Z(\mathcal{A}(B)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

If, in addition, the reference state on $A \cup B \cup D$ is exact Markov,

$$I(A : D|B)_\rho = 0,$$

then the state decomposes as

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha\, \rho^{(\alpha)}_{Ab_L^\alpha} \otimes \rho^{(\alpha)}_{b_R^\alpha D}.$$

The same formula holds in any limit state that is exact Markov on this fixed collar model. Thus the collar-center block decomposition is forced by overlap consistency plus the derived regulator package on the ordinary or central-defect branch; exact Markov factorization is an additional state condition, not a consequence of the block decomposition alone.

Proof. By Proposition 5.1, one may realize the physical collar states as the $\widehat{K}_\Sigma$-invariant subspace of $\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R}$. Complete reducibility of finite-dimensional unitary representations gives the displayed

decomposition of $\tilde{\mathcal{H}}_{B_L}$, and the right half-collar carries the inverse transport law across the same cut, hence the contragredient decomposition of $\tilde{\mathcal{H}}_{B_R}$. Therefore

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*) \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

Taking $\widehat{K}_\Sigma$-invariants gives

$$\mathcal{H}_B \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

By Schur's lemma,

$$(W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

which yields the displayed block decomposition and shows that the center is generated by the block projectors. Within each block, observables from $A \cup B$ act only on the left factor and observables from $B \cup D$ act only on the right factor. If the state is exact Markov, the standard HJPW structure theorem gives the blockwise factorized normal form [**?**]. The same formula is used in the idealized recoverability limit when exact identities are taken literally. □

Remark 5.3 (Status of the regulator package and the precise Markov boundary)**.** *Proposition 5.1 and Theorem 5.2 show that, on the ordinary or central-defect branch, edge-center completion is a theorem of overlap consistency plus the finite regulator presentation; no preferred quantum-link realization is being assumed there. Assumption 3.6**T1** is therefore theorem-local shorthand for this derived regulator package. The only remaining fixed-cutoff obstruction to a fully general ordinary-group theorem is a genuinely noncentral 2-group defect. On that branch the fixed-cutoff topological UV package is therefore closed up to that named noncentral obstruction; the further transportable compact-gauge lift still carries **T7** and **T10**–**T12** whenever it is invoked. The distinct issue addressed next is* state *control: exact splice and modular-additivity identities require either literal exact Markovity or a controlled family that converges to the exact Markov set on one fixed finite-dimensional collar.*

Definition 5.4 (Exact Markov set and collar-distance modulus)**.** *Fix the finite-dimensional collar Hilbert space of Theorem 5.2. Let*

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : \ I(A:D \mid B)_\sigma = 0\}$$

be the set of exact Markov states on that fixed collar. Equivalently, by HJPW and Theorem 5.2, $\sigma \in \mathfrak{M}_{A:B:D}$ iff

$$\sigma_{ABD} = \bigoplus_\alpha q_\alpha \, \sigma^{(\alpha)}_{Ab_L^\alpha} \otimes \sigma^{(\alpha)}_{b_R^\alpha D}$$

for the fixed edge-center decomposition of B.

For a general state ρ_{ABD}, define its distance to the exact Markov set by

$$d_{\mathrm{M}}(\rho) := \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1,$$

and the fixed-collar exact-Markov modulus by

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup \{ d_{\mathrm{M}}(\rho) : I(A:D \mid B)_\rho \leq \varepsilon \}.$$

Proposition 5.5 (Controlled exact Markov-collar limit)**.** *On a fixed finite-dimensional collar, the exact Markov set $\mathfrak{M}_{A:B:D}$ is compact and*

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad \text{as } \varepsilon \downarrow 0.$$

Consequently, if $\rho^{(n)}_{ABD}$ is any sequence of states on that same collar with

$$I(A:D \mid B)_{\rho^{(n)}} \le \varepsilon_n, \qquad \varepsilon_n \to 0,$$

then there exist exact Markov states $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ such that

$$\|\rho^{(n)} - \sigma^{(n)}\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) \longrightarrow 0.$$

This is the precise sense in which approximate recoverability converges to the exact HJPW normal form used later.

Proof. The full state space on a fixed finite-dimensional Hilbert space is compact in trace norm. Conditional mutual information is continuous there because von Neumann entropy is continuous in finite dimension. Hence $\mathfrak{M}_{A:B:D}$ is closed, so it is compact as a closed subset of a compact space.

Suppose $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \not\to 0$. Then there exist $\eta > 0$, a sequence $\varepsilon_n \downarrow 0$, and states $\rho^{(n)}$ with $I(A:D \mid B)_{\rho^{(n)}} \le \varepsilon_n$ but $d_{\mathrm{M}}(\rho^{(n)}) \ge \eta$ for all n. By compactness, pass to a trace-norm convergent subsequence $\rho^{(n_k)} \to \rho^*$. Continuity of conditional mutual information gives

$$I(A:D \mid B)_{\rho^*} = \lim_{k\to\infty} I(A:D \mid B)_{\rho^{(n_k)}} = 0,$$

so $\rho^* \in \mathfrak{M}_{A:B:D}$. But then

$$d_{\mathrm{M}}(\rho^{(n_k)}) \le \|\rho^{(n_k)} - \rho^*\|_1 \longrightarrow 0,$$

contradicting $d_{\mathrm{M}}(\rho^{(n_k)}) \ge \eta$. Thus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$. The final statement follows by choosing $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ within $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) + 1/n$ of $\rho^{(n)}$. □

Proposition 5.6 (Approximate collar recovery)**.** *Let A-B-D be a collar tripartition satisfying*

$$I(A:D|B)_\rho \le \varepsilon,$$

and let $\mathcal{R}_{B\to BD}$ be a recovery map such that

$$F(\rho_{ABD}, (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})) \ge e^{-\varepsilon/2}.$$

Define the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABD} := (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB}), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Then

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le r_{\mathrm{FR}}(\varepsilon).$$

Consequently, for every CPTP map $\Lambda_{BD\to B'D'}$ and every bounded observable X supported on $A\cup B'$,

$$|\mathrm{Tr}[X(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})]| \le \|X\|_\infty \, \|(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})\|_1 \le \|X\|_\infty \, r_{\mathrm{FR}}(\varepsilon).$$

Proof. The fidelity lower bound and the Fuchs–van de Graaf inequality give

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le 2\sqrt{1 - F(\rho_{ABD}, \rho^{\mathrm{rec}}_{ABD})^2} \le 2\sqrt{1 - e^{-\varepsilon}} = r_{\mathrm{FR}}(\varepsilon).$$

The estimate $r_{\mathrm{FR}}(\varepsilon) \le 2\sqrt{\varepsilon}$ follows from $1 - e^{-x} \le x$. Contractivity of trace norm under CPTP maps gives the transported bound, and the observable estimate is the duality between trace norm and operator norm [**?**]. The bound is constructive and dimension-free. What it controls is closeness to a recovered comparison state; no claim is made that $\rho^{\mathrm{rec}}_{ABD}$ is itself exact Markov. □

Proposition 5.7 (Exact splice theorem and controlled collar approximation)**.** *Under Theorem 5.2, let $\sigma_{ABD} \in \mathfrak{M}_{A:B:D}$ have exact Markov form*

$$\sigma_{ABD} = \bigoplus_\alpha p_\alpha\, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D}.$$

Let $\tau^{(\alpha)}_{b^\alpha_R D'}$ be any normalized family of states compatible with the same right-boundary sectors, and define

$$\sigma'_{ABD'} = \bigoplus_\alpha p_\alpha\, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \tau^{(\alpha)}_{b^\alpha_R D'}.$$

Define the common left algebra

$$\mathcal{A}_{Ab_L} := \bigoplus_\alpha \mathcal{B}(\mathcal{H}_{Ab^\alpha_L}) \otimes \mathbf{1}_{b^\alpha_R},$$

canonically represented on both direct-sum Hilbert spaces. Then:

1. *for every $X \in \mathcal{A}_{Ab_L}$,*
$$\mathrm{Tr}(X\sigma'_{ABD'}) = \mathrm{Tr}(X\sigma_{ABD});$$
2. *if ρ_{ABD} is any state on the same fixed collar with $I(A:D \mid B)_\rho \le \varepsilon$, and if $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ is chosen so that*
$$\|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon),$$
then the corresponding exact splice σ'_ε satisfies
$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$
for all $X \in \mathcal{A}_{Ab_L}$.

Hence the exact splice identity used later is justified either at exact Markovity or along any controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta) \to 0$.

Proof. For item 1, blockwise factorization gives

$$\mathrm{Tr}(X\sigma'_{ABD'}) = \sum_\alpha p_\alpha \mathrm{Tr}\Big(X_\alpha \sigma^{(\alpha)}_{Ab^\alpha_L}\Big)\, \mathrm{Tr}\Big(\tau^{(\alpha)}_{b^\alpha_R D'}\Big).$$

Here X_α denotes the α-block of X in $\mathcal{A}_{Ab_L}$. Each $\tau^{(\alpha)}$ is normalized, so the second factor is 1. The same computation for σ_{ABD} gives the same value because the original right factor is likewise normalized.

For item 2, apply item 1 to σ_ε:

$$\mathrm{Tr}(X\sigma'_\varepsilon) = \mathrm{Tr}(X\sigma_\varepsilon).$$

Therefore

$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| = \left|\mathrm{Tr}[X(\rho_{ABD} - \sigma_\varepsilon)]\right| \le \|X\|_\infty\, \|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \|X\|_\infty\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

□

Proposition 5.8 (Transport of modular-additivity errors from the exact Markov reference)**.** *Let $K_X(\omega) := -\log \omega_X$ for a faithful reduced state on region X, and define the modular defect*

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Fix a collar and let ρ be a faithful state with $I(A : D \mid B)_\rho \le \varepsilon$. Choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Assume that the four marginals ρ_X and $(\sigma_\varepsilon)_X$ for $X \in \{ABD, AB, BD, B\}$ all satisfy

$$\rho_X \ge \lambda_* \mathbf{1}, \qquad (\sigma_\varepsilon)_X \ge \lambda_* \mathbf{1}$$

on their supports for some $\lambda_* > 0$. Then

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Since $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, every matrix element of the modular defect of ρ is within $4\lambda_*^{-1}\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ of the exact Markov value.

Proof. By monotonicity of trace distance under partial trace,

$$\|\rho_X - (\sigma_\varepsilon)_X\|_1 \le \|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$

for each $X \in \{ABD, AB, BD, B\}$. On the interval $[\lambda_*, 1]$, the function $\log x$ has derivative bounded by λ_*^{-1}. By the standard integral representation of the operator logarithm, this implies the operator-norm Lipschitz bound

$$\|K_X(\rho) - K_X(\sigma_\varepsilon)\|_\infty = \|\log(\sigma_\varepsilon)_X - \log \rho_X\|_\infty \le \lambda_*^{-1}\, \|\rho_X - (\sigma_\varepsilon)_X\|_\infty \le \lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Summing the four region contributions gives

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

For $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$, the exact HJPW block factorization implies that $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, so the same bound controls all matrix elements relative to the exact Markov modular-additivity value. □

Remark 5.9 (What is and is not carried forward)**.** *There are therefore two distinct quantitative controls, and the compact note keeps them separate:*

1. *the constructive one-shot recoverability bound $r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2})$, which compares the physical state to a recovered comparison state and is enough for bounded-observable statements at one regulator stage;*

2. *the fixed-collar exact-Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$, which compares the physical state to the exact Markov set and is the correct quantity for justifying exact splice or modular-additivity identities in a controlled limit, with an additional factor $4\lambda_*^{-1}$ when logarithms are involved.*

Later spatial and null collar theorems therefore use exact identities only in two regimes: literal exact Markovity, or a controlled refinement or scaling family in which the relevant $\delta^{\mathrm{M}}(\varepsilon_\delta)$ tends to zero after transport to one fixed finite-dimensional collar model. The manuscript does not *claim a universal dimension-free one-shot trace-norm estimate from small conditional mutual information directly to an exact Markov state.*

Proposition 5.10 (Generalized-entropy split)**.** *If the reduced cap state takes the block form*

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha} \right),$$

then

$$S(\rho_C) = S_{\mathrm{bulk}}(C) + \mathrm{Tr}(\rho_C L_C),$$

where

$$S_{\mathrm{bulk}}(C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^{(\alpha)}_{\mathrm{bulk},C}), \qquad L_C = \sum_\alpha (\log d_\alpha) P_\alpha.$$

Proof. The entropy of a direct sum is

$$S\left(\bigoplus_\alpha p_\alpha \sigma_\alpha \right) = H(p_\alpha) + \sum_\alpha p_\alpha S(\sigma_\alpha).$$

Apply this identity with

$$\sigma_\alpha = \rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha}.$$

Since

$$S(\sigma_\alpha) = S(\rho^{(\alpha)}_{\mathrm{bulk},C}) + \log d_\alpha,$$

the first statement follows. □

Definition 5.11 (Effective Newton constant in the refinement dictionary)**.** *In the refinement-scaling regime, if*

$$\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t), \qquad A(\partial C) \approx N_\Sigma a_{\mathrm{cell}},$$

then matching $\mathrm{Tr}(\rho_C L_C)$ *to the area term* $A(\partial C)/(4G)$ *defines the effective Newton constant*

$$G := \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

This is a continuum dictionary identification, not part of the exact algebraic proposition.

Proposition 5.7 is the algebraic version of interior invariance under compatible exterior substitutions. Proposition 5.8 is the precise bridge from small collar conditional mutual information to the exact modular identities later used by the spatial and null branches: at finite stage one carries a remainder, and exact identities appear only when that remainder is zero or tends to zero in the controlled collar limit. Proposition 5.10 isolates the exact entropy split, while Definition 5.11 supplies the continuum identification that turns the edge center into a gravitational coupling.

6 Modular Geometry, Relativity, and Einstein Dynamics

6.1 Geometric modular flow

The collar analysis identifies where modular nonadditivity can live, but by itself it does not determine the continuum geometric action of the limiting modular group. The theorem therefore works on the refinement-stable cap branch lying in the Bisognano–Wichmann geometric modular class of **T4**. On that branch, OPH supplies a controlled reduction from the regulator collar to the

refinement limit together with the 2π normalization and carried-error statement for the standard cap-preserving conformal subgroup.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C, with $\delta \downarrow 0$ and $A_\delta \cup B_\delta \cup D_\delta$ covering a neighborhood of the cut. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. By the collar results proved earlier, all later "exact" collar formulas are to be read in one of two ways:

1. literal exactness when the reference collar state is exact Markov; or
2. a controlled collar family for which

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with the finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

Theorem 6.1 (Geometric modular flow on caps in the OPH Bisognano–Wichmann branch)**.** *Assume Axioms 3.1–3.4, together with the fixed-cutoff regulator presentation **T1**, the branch-internal control statements denoted **T2**–**T3**, and the controlled scaling-limit scope clause **T8** of Assumption 3.6. Assume in addition the **T4** branch condition that the refinement-stable MaxEnt branch selected by OPH lies in the Bisognano–Wichmann geometric modular class for caps and their conformal images. For each cap $C \subset S^2$, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup supplied by **T4**, with geometric parameter s normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let B_C be its infinitesimal generator. Then for every cap $C \subset S^2$ the following hold.*

(i) *At regulator scale, the modular defect is confined to the shrinking collar up to the carried errors measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}.*

(ii) *In the refinement limit, the modular action on $\mathcal{A}(C)$ is geometric and is implemented by $\lambda_C(\kappa_C t)$ for some $\kappa_C > 0$. Equivalently, the limiting modular Hamiltonian is of the form*

$$K_C = \kappa_C\, B_C.$$

(iii) *The modular KMS condition fixes the normalization to*

$$\kappa_C = 2\pi.$$

Consequently, for every controlled collar family with

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$$

and surviving faithfulness bound, there exists a carried remainder $E_C^{(\delta)}$ such that

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)},$$

where the effect of $E_C^{(\delta)}$ on bounded observables is controlled by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and its modular-Hamiltonian replacement error is controlled by η_δ^{M}. In the exact-Markov case one may take $E_C^{(\delta)} = 0$ at that collar scale; in the controlled refinement limit,

$$K_C = 2\pi B_C.$$

Proof. Step 1: collar localization. At one fixed collar scale, Propositions 5.7 and 5.10 together with the controlled Markov replacement results show that all exact collar identities are justified either literally at exact Markovity or along a family whose failure of exactness is measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. In particular, the nonadditive part of the modular generator is confined to the shrinking collar around ∂C up to those carried errors.

Step 2: passage to the Bisognano–Wichmann geometric branch. By the **T4** branch hypothesis, the refinement-stable cap net belongs to the Bisognano–Wichmann geometric modular class. Therefore the limiting modular automorphism group of $\mathcal{A}(C)$ is implemented by the standard cap-preserving conformal one-parameter subgroup $\lambda_C(s)$. Writing B_C for the infinitesimal generator supplied by that branch data, there exists $\kappa_C > 0$ such that

$$K_C = \kappa_C B_C.$$

Step 3: normalization. The modular flow is KMS at inverse temperature 1 by definition. Since the geometric parameter s in $\lambda_C(s)$ is normalized by unit null dilation in the **T4** branch, the usual Bisognano–Wichmann relation identifies the modular parameter with the geometric one by $s = 2\pi t$. Equivalently,

$$\kappa_C = 2\pi.$$

Hence the null blow-up near a smooth cut acts by $v \mapsto e^{-2\pi t} v$.

Step 4: controlled limit. At finite collar scale the only discrepancy between the physical modular generator and its exact geometric normal form is the carried collar-replacement error already isolated above. Hence one may write

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)}$$

with bounded-observable and modular-Hamiltonian effects controlled respectively by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. When the collar family satisfies

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$$

and the lower spectral bound survives, the carried remainder vanishes in the refinement limit, yielding

$$K_C = 2\pi B_C.$$

□

Remark 6.2 (What is proved and what is not)**.** *The theorem proves the controlled collar reduction together with the* 2π *normalization* on the OPH refinement branch in which the cap net has the Bisognano–Wichmann geometric modular action of **T4**. *On that branch, the standard cap-preserving conformal subgroup is part of the branch data, and the theorem identifies its modular generator with the carried collar remainder made explicit. The branch-selection question is encoded in the stated **T4** hypothesis; the theorem then fixes the normalization and controlled refinement limit on that branch.*

Corollary 6.3 (Lorentz Kinematics)**.** *Under the hypotheses of Theorem 6.1,*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The cap modular flows are therefore the standard one-parameter Lorentz boost/dilation subgroups in the celestial-sphere realization, and the induced local kinematic group is the connected Lorentz group.

Proof. Orientation-preserving conformal maps of S^2 are exactly the Möbius transformations, so

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

By Theorem 6.1 together with the **T4** branch hypothesis, the cap modular groups are identified with the standard cap-preserving conformal dilations inside this group. Hence the local kinematics induced by modular flow is the connected Lorentz group. □

6.2 The null modular bridge

The fixed-cutoff strip analysis begins from the transferred cut-center data and the inherited-strip Markov structure. At fixed regulator scale one obtains exact or controlled four-term strip additivity together with endpoint-Lipschitz control of the renormalized half-line family; on the same OPH geometric scaling branch used in Theorem 6.1, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion. Standard Borchers–Wiesbrock theory consequently applies to the resulting OPH half-sided modular pair, yielding positivity and unitary implementation of null translations. What is still not proved here is the downstream density upgrade from the weak tail generator to a local operator-valued density and the relativistic identification of that density with continuum T_{kk}.

The present subsection therefore establishes the following chain:

1. null-strip center transfer and the inherited left/right split hypothesis;
2. exact strip additivity on the exact inherited Markov model, and a carried defect operator on controlled exact-Markov replacements on one fixed strip model;
3. endpoint-Lipschitz matrix elements for the renormalized half-line family and the resulting weak tail generator;
4. derived half-sided modular inclusion on the null half-line blow-up net from OPH scaling-limit geometry, and the resulting Borchers positive translation generator.

The labels 6.8 and 6.11 are retained below as the only explicit downstream boundaries. Historical references elsewhere to the old null half-sided-inclusion label **T6** should, within this subsection, be read as shorthand for the derived statement of Corollary 6.9. The positive null-translation generator is now stated explicitly in Lemma 6.10: the Borchers unitary group is constructed on its Stone domain, positivity is theorem-level rather than implicit, and the half-line modular Hamiltonians are tied to it by explicit affine covariance and quadratic-form identities. Bounded-interval formulas remain downstream and continue to require the separate interval-preserving projective branch.

Proposition 6.4 (Null-strip center transfer and inherited split)**.** *Fix a null generator Ω and a regulated tripartition*

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts

$$\Gamma_- := \{v = v_2\}, \qquad \Gamma_+ := \{v = v_3\}.$$

*Assume the fixed-cutoff regulator package **T1** of Assumption 3.6, and assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data of Proposition 5.1 in the following precise sense:*

(i) each cut $\Gamma_\pm$ carries a compact derived boundary action $\widehat{K}_\pm$ on a reference cut Hilbert space, with $\widehat{K}_\pm = K_\pm$ on the ordinary branch;

(ii) in a compatible type-I regulator presentation, the adjacent regions carry inverse transport across each cut, so for irreducible $\widehat{K}_\pm$-modules $W_{\alpha_\pm}$ one has decompositions

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}};$$

(iii) the gauge-invariant strip algebra is the commutant of the transported boundary action on $\tilde{\mathcal{H}}_J$, so that the pair (α_-, α_+) is a central cut label;

(iv) for each (α_-, α_+), the multiplicity space $\mathcal{H}_{j^{\alpha_-,\alpha_+}}$ admits a chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

such that, blockwise, $\mathcal{A}(I_- \cup J)$ acts only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acts only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$.

Then the strip algebra has the inherited central decomposition

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+},$$

and, under item (iv),

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}).$$

Moreover the glued tripartition carries the block decomposition

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Thus the two null cuts transfer the same center data as the spatial collar branch, while the left/right split of the strip multiplicity spaces is exactly the extra decomposition-inheritance hypothesis and is not forced by center transfer alone.

Proof. By complete reducibility of the finite-dimensional unitary actions $\widehat{K}_\pm$, the displayed decompositions of $\tilde{\mathcal{H}}_{I_-}$, $\tilde{\mathcal{H}}_J$, and $\tilde{\mathcal{H}}_{I_+}$ exist. Under item (iii), the strip algebra is the commutant of the transported $\widehat{K}_- \times \widehat{K}_+$ action on

$$\tilde{\mathcal{H}}_J = \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+}.$$

Write an operator $X \in \mathcal{B}(\tilde{\mathcal{H}}_J)$ in matrix form relative to that direct sum. The block

$$X_{(\alpha_-,\alpha_+),(\beta_-,\beta_+)}$$

intertwines

$$W^*_{\alpha_-} \otimes W_{\alpha_+} \longrightarrow W^*_{\beta_-} \otimes W_{\beta_+}$$

for the product action of $\widehat{K}_- \times \widehat{K}_+$. By Schur's lemma, such an intertwiner is zero unless $(\alpha_-, \alpha_+) = (\beta_-, \beta_+)$, and when the sector pair agrees it is scalar on the representation factors. Therefore the commutant is exactly

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}),$$

and the center is generated by the corresponding block projectors P_{α_-,α_+}.

For the glued tripartition, tensor the three regulator Hilbert spaces and decompose:

$$\tilde{\mathcal{H}}_{I_-} \otimes \tilde{\mathcal{H}}_J \otimes \tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_-,\beta_-,\alpha_+,\beta_+} (W_{\alpha_-} \otimes W^*_{\beta_-}) \otimes \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\beta_-,\beta_+}} \otimes (W_{\beta_+} \otimes W^*_{\alpha_+}) \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Taking $\widehat{K}_- \times \widehat{K}_+$-invariants and applying Schur's lemma to each cut gives

$$(W_{\alpha_-} \otimes W^*_{\beta_-})^{\widehat{K}_-} \cong \begin{cases} \mathbb{C}, & \alpha_- = \beta_-, \\ 0, & \alpha_- \neq \beta_-, \end{cases} \qquad (W_{\beta_+} \otimes W^*_{\alpha_+})^{\widehat{K}_+} \cong \begin{cases} \mathbb{C}, & \beta_+ = \alpha_+, \\ 0, & \beta_+ \neq \alpha_+, \end{cases}$$

so only matching cut sectors survive and one obtains

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

If item (iv) holds, substitute the chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

to obtain the displayed inherited split and the stated action of the left and right union algebras. Nothing in the Schur-lemma argument itself forces item (iv); it is exactly the extra structural condition required to reproduce the same blockwise tensor pattern as Theorem 5.2. □

Corollary 6.5 (Exact or controlled four-term null modular relation on an inherited strip model)**.** *Fix one finite-dimensional regulated strip model satisfying Proposition 6.4, and define*

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

For any state η on this strip model, write

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the reference state ω is exact Markov on this inherited decomposition, then

$$\Delta K_J(\omega) \in Z(\mathcal{A}(J)),$$

and in the canonical blockwise identification of Proposition 6.4 one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, without fixing that normalization there exists a central operator

$$K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$$

such that

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, then on this fixed strip model one may choose an exact Markov state $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

where

$$K_{\partial,J}(\widetilde{\omega}_J) := \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J)).$$

For every bounded observable O in $\mathcal{A}(I_- \cup J)$ or $\mathcal{A}(J \cup I_+)$,

$$\big|\omega(O) - \widetilde{\omega}_J(O)\big| \le \|O\|_\infty \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the marginals entering ΔK_J for ω and $\widetilde{\omega}_J$ are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, the Fawzi–Renner recovery map supplies a constructive comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Hence the exact four-term strip relation is available only at exact Markovity, or along a controlled strip family on one fixed inherited strip model for which

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0.$$

In the controlled case, $\mathfrak{D}_J$ is carried explicitly at each finite stage.

Proof. Assume first that ω is exact Markov on the inherited split of Proposition 6.4. Then the HJPW structure theorem gives, on each sector pair $\alpha = (\alpha_-, \alpha_+)$,

$$\omega_{I_- J I_+}|_\alpha = p_\alpha \, \omega^{(\alpha)}_{I_- j_L} \otimes \omega^{(\alpha)}_{j_R I_+}.$$

Accordingly,

$$K_{I_-\cup J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R I_+}.$$

Tracing over I_+, I_-, or both gives the marginal block formulas

$$K_{I_-\cup J}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R},$$

$$K_{J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R I_+},$$

$$K_J(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R}.$$

Subtracting the last three expressions from the first gives zero on every block. Thus, in the canonical blockwise identification supplied by Proposition 6.4,

$$\Delta K_J(\omega) = 0.$$

If one chooses to keep the blockwise endpoint-label bookkeeping explicit rather than absorbing those constants into the canonical identification, the remainder is a direct sum of block constants and is therefore central. This yields the stated central form with $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$.

For the controlled statement, apply Proposition 5.5 on this fixed finite-dimensional strip model, relabeling A, B, D there as I_-, J, I_+. This yields an exact Markov replacement $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with the displayed trace-norm bound. The identity

$$\Delta K_J(\omega) = \Delta K_J(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J)$$

is tautological, so the displayed four-term formula follows from the exact-Markov case applied to $\widetilde{\omega}_J$. The observable estimate is immediate from

$$\left|\omega(O) - \widetilde{\omega}_J(O)\right| \leq \|O\|_\infty \, \|\omega - \widetilde{\omega}_J\|_1.$$

Finally, Proposition 5.8, again relabeled $A, B, D \mapsto I_-, J, I_+$, yields the operator-norm bound on $\mathfrak{D}_J$ whenever the relevant marginals have the stated lower spectral bound. The Fawzi–Renner estimate is the same one-shot recoverability bound already used elsewhere and is independent of the exact-Markov replacement modulus. □

Definition 6.6 (Renormalized null modular functional)**.** *For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 6.5 on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define*

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega).$$

For a null half-line $H_a := (a, \infty)$, abbreviate

$$\widetilde{K}_a(\Omega) := \widetilde{K}[H_a, \Omega].$$

Proposition 6.7 (Endpoint-Lipschitz null modular families and weak tail generator)**.** *Under the derived quasi-local propagation and bounded-interval endpoint control already internal to Axiom 3.3, the renormalized interval family obeys the following matrix-element bounds on every fixed local-energy-bounded domain. For bounded intervals*

$$I = (a, b), \qquad I' = (a', b'),$$

contained in one compact endpoint window,

$$\left|\langle \psi, (\widetilde{K}[I', \Omega] - \widetilde{K}[I, \Omega])\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \left(|a' - a| + |b' - b|\right).$$

In particular the matrix elements of $\widetilde{K}[I, \Omega]$ are jointly Lipschitz in the two endpoints and therefore have finite variation there.

For null half-lines $H_a = (a, \infty)$, one likewise has

$$\left|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \, |a' - a|$$

for a, a' in every compact window. Hence, for fixed ψ, ϕ,

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous. Its distributional derivative defines a locally L^∞ sesquilinear-form density

$$\langle \psi, q(a,\Omega)\phi \rangle := -\frac{1}{2\pi}\, \partial_a f_{\psi,\phi}(a),$$

and for $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi \rangle \, dv.$$

The object $q(a,\Omega)$ is the weak tail generator. At this stage it is not yet asserted to be represented by a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after the derived half-sided modular inclusion of Corollary 6.9 and Lemma 6.10.

Proof. The bounded-interval endpoint-control statement already derived from Axiom 3.3 applies precisely to the renormalized family obtained after removal of the central endpoint term. Therefore, for intervals contained in one fixed compact endpoint window,

$$\left| \langle \psi, (\widetilde{K}[I',\Omega] - \widetilde{K}[I,\Omega])\phi \rangle \right| \le C_{\psi,\phi,\Omega}\, |I' \Delta I|,$$

where $I'\Delta I$ is the symmetric difference of the two intervals. If

$$I = (a,b), \qquad I' = (a',b'),$$

then

$$|I'\Delta I| \le |a' - a| + |b' - b|,$$

which gives the displayed joint endpoint-Lipschitz bound.

For half-lines $H_a = (a,\infty)$ and $H_{a'} = (a',\infty)$, the symmetric difference is the bounded interval between the two endpoints, so

$$|H_{a'} \Delta H_a| = |a' - a|.$$

Because the central endpoint-label term has already been removed in the definition of $\widetilde{K}_a$, the same endpoint-control estimate yields

$$\left| \langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle \right| \le C_{\psi,\phi,\Omega}\, |a' - a|.$$

Hence, on every compact a-interval, the function

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is Lipschitz. A Lipschitz function on a compact interval is absolutely continuous, so it has an almost-everywhere derivative in L^∞ and obeys the fundamental theorem of calculus:

$$f_{\psi,\phi}(b) - f_{\psi,\phi}(a) = \int_a^b \partial_v f_{\psi,\phi}(v)\, dv.$$

Define

$$\langle \psi, q(v,\Omega)\phi \rangle := -\frac{1}{2\pi}\, \partial_v f_{\psi,\phi}(v)$$

distributionally. Substituting this definition into the previous identity gives

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi \rangle \, dv.$$

This is exactly the claimed weak-tail formula. □

Theorem 6.8 (Downstream density-upgrade template)**.** *Fix a null generator Ω and the renormalized half-line family $\widetilde{K}_a(\Omega)$ of Proposition 6.7. For fixed vectors ψ, ϕ in a common dense domain, let*

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle, \qquad q_{\psi,\phi}(a) := \langle \psi, q(a,\Omega)\phi \rangle = -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

Assume in addition that $q_{\psi,\phi}$ is weakly differentiable in a, that

$$\lim_{a\to+\infty} f_{\psi,\phi}(a) = 0, \qquad \lim_{a\to+\infty} q_{\psi,\phi}(a) = 0,$$

and define distributionally

$$\langle \psi, p(a,\Omega)\phi \rangle := -\partial_a q_{\psi,\phi}(a) = \frac{1}{2\pi}\,\partial_a^2 f_{\psi,\phi}(a).$$

Then

$$q_{\psi,\phi}(a) = \int_a^\infty \langle \psi, p(v,\Omega)\phi \rangle\, dv,$$

and

$$\langle \psi, \widetilde{K}_a(\Omega)\phi \rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi \rangle\, dv.$$

This statement is not part of the present fixed-cutoff bridge; it records the exact additional input/output package carried into the downstream density-upgrade task.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a, \infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

Thus $\mathcal{M}_a(\Omega)$ is the half-line blow-up algebra generated by bounded null interval algebras inside H_a. By isotony of the null interval net,

$$a \leq b \quad \Longrightarrow \quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 6.9 (Derived half-sided modular inclusion on null half-lines)**.** *On the OPH geometric scaling branch of Theorem 6.1, the blow-up modular action near a smooth entangling cut acts on the null coordinate by*

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \leq 0),$$

so the inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$$

is half-sided modular. Equivalently, after the harmless convention change $t \mapsto -t$, one recovers the standard positive-time half-sided-inclusion form. The former half-sided-inclusion premise is therefore derived from OPH null-interval structure, isotony, and the scaling-limit geometric action rather than imported as an additional modular-QFT ingredient.

Proof. By Theorem 6.1, on every controlled collar family whose carried remainder vanishes in the refinement limit, the cap modular flow converges to the exact geometric cap dilation with 2π normalization. Blow up the cap near a smooth cut and restrict to the chosen null generator. In that tangent limit, the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, the blow-up action therefore sends

$$\mathcal{A}((c, d), \Omega) \longmapsto \mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega).$$

Taking the von Neumann closure of the algebra generated by all such intervals gives

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

If $t \leq 0$, then $e^{-2\pi t}a \geq a$, hence

$$H_{e^{-2\pi t}a} \subseteq H_a.$$

Isotony of the null interval net therefore implies

$$\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

This is exactly the half-sided modular-inclusion property for the pair $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$. Reparametrizing the modular parameter by $t \mapsto -t$ gives the more common positive-time convention if desired. □

Lemma 6.10 (Positive null-translation generator)**.** *For the derived half-sided modular inclusion*

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0)$$

of Corollary 6.9, let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define

$$K_0(\Omega) := -\log \Delta_0(\Omega).$$

Then Borchers–Wiesbrock yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\,\mathcal{M}_b(\Omega)\,U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{\psi \in \mathcal{H} : \lim_{a\to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists}\right\}.$$

Moreover,

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad \Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

For each $a \geq 0$, let $\Delta_a(\Omega)$ be the modular operator of the translated standard pair $(\mathcal{M}_a(\Omega), \omega)$ and define

$$K_a(\Omega) := -\log \Delta_a(\Omega).$$

Then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

hence

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$*. Equivalently,*

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi\rangle = -2\pi(b-a)\,\langle \psi, P_\Omega\phi\rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$*.*

In the canonical blockwise normalization of Corollary 6.5, where the half-line endpoint term has been absorbed into the central part, the weak endpoint derivative of Proposition 6.7 agrees with the Borchers generator:

$$\langle \psi, P_\Omega\phi\rangle = -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+}\langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

for the same domain vectors. The half-sided-inclusion step proves only this affine half-line pair $(K_0(\Omega), P_\Omega)$*; bounded-interval modular-Hamiltonian formulas remain downstream consequences of the separate interval-preserving projective branch recorded in Theorem 6.11.*

Proof. Corollary 6.9 gives the required half-sided modular inclusion. The standard Borchers–Wiesbrock theorem for a standard half-sided inclusion then yields a unique strongly continuous unitary group $U_\Omega(a)$ with $U_\Omega(a)\omega = \omega$, the translation law for the nested half-line net, and a positive self-adjoint generator P_Ω; Stone's theorem gives the displayed domain formula.

The affine covariance relation

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a)$$

is the Borchers relation. Differentiating at $a = 0$ gives

$$\Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega.$$

Since $\Delta_0(\Omega)^{it} = e^{-itK_0(\Omega)}$, differentiating at $t = 0$ on the common analytic core yields

$$-i[K_0(\Omega), P_\Omega] = -2\pi P_\Omega,$$

hence

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and

$$\mathcal{M}_a(\Omega) = U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^*,$$

the modular operator of the translated standard pair is

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*.$$

Differentiating this identity in a on $D(K_0(\Omega)) \cap D(P_\Omega)$ gives

$$\frac{d}{da}K_a(\Omega) = i[P_\Omega, K_a(\Omega)] = -2\pi P_\Omega,$$

because the commutator relation just proved is invariant under conjugation by $U_\Omega(a)$. Integrating from 0 to a yields the displayed quadratic-form identity, and taking matrix elements gives the equivalent form. Proposition 6.7 had already produced the weak endpoint derivative of the renormalized half-line family; in the canonical normalization of Corollary 6.5 the central term has been removed, so that weak derivative is exactly the Borchers generator. The final sentence is the explicit claim boundary: bounded intervals require the additional projective interval action not supplied by half-sided inclusion alone. □

Theorem 6.11 (Downstream null-stress identification template)**.** *Assume Axioms 3.1–3.4, Assumption 3.6, Proposition 6.4, the exact-or-controlled strip hypothesis of Corollary 6.5, Proposition 6.7, the density-upgrade hypotheses of Theorem 6.8, and the scaling-limit null-stress identification premise of the relativistic EFT branch. Then, on the half-line blow-up net whose half-sided modular inclusion is derived in Corollary 6.9 and whose positive null-translation generator is identified in Lemma 6.10, there exists an operator-valued distribution $T_{kk}(v,\Omega)$ such that the renormalized modular Hamiltonian on a null half-line takes the form*

$$\widetilde{K}[(a,\infty),\Omega] = 2\pi \int_a^\infty (v-a)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{(a,\infty),\Omega},$$

where for each relevant half-line or bounded interval I, ε_I denotes the strip-CMI control parameter and δ_I^{M} the corresponding fixed-strip exact-Markov modulus of Corollary 6.5, and

$$\eta_I := r_{\mathrm{FR}}(\varepsilon_I) + 4\lambda_*^{-1}\delta_I^{\mathrm{M}}(\varepsilon_I) + \eta_I^{\mathrm{EFT}},$$

and for all domain vectors ψ, ϕ,

$$|\langle \psi, E^{(\eta)}_{I,\Omega}\phi\rangle| \le C_{I,\Omega,\psi,\phi}\, \eta_I.$$

If bounded null intervals inherit geometric modular action from the interval-preserving projective flow, then for each bounded interval I one likewise has

$$\widetilde{K}[I,\Omega] = 2\pi \int_I g_I(v)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{I,\Omega},$$

where $g_I(v)$ is affine-covariant and vanishes at finite interval endpoints. The full modular Hamiltonian is then

$$K[I,\Omega] = \widetilde{K}[I,\Omega] + K_\partial(I,\Omega),$$

with K_∂ central and boundary-supported. Knowledge of T_{kk} for all null directions reconstructs a symmetric tensor T_{ab} up to the addition of a metric term ϕg_{ab}. This theorem is not yet established at the present c.12 step; what remains beyond the derived half-sided-inclusion bridge is precisely the density upgrade of Theorem 6.8 and the relativistic identification of that density with continuum null stress.

Lemma 6.12 (Null data determine the stress tensor modulo a metric term)**.** *Let X_{ab} be a symmetric tensor. If*

$$X_{ab}k^a k^b = 0$$

for every null vector k at a point, then

$$X_{ab} = \phi g_{ab}$$

for some scalar ϕ.

Proof. Write

$$X_{ab} = Y_{ab} + \phi g_{ab}$$

with Y_{ab} traceless. Since $g_{ab}k^a k^b = 0$ on null vectors, the hypothesis implies $Y_{ab}k^a k^b = 0$ for all null k. In local inertial coordinates let $k = (1, \hat{n})$ with $|\hat{n}| = 1$. Then

$$Y_{ab}k^a k^b = Y_{00} + 2\hat{n}^i Y_{0i} + \hat{n}^i \hat{n}^j Y_{ij}.$$

Oddness under $\hat{n} \mapsto -\hat{n}$ forces $Y_{0i} = 0$. Averaging the remaining relation over S^2 gives

$$Y_{00} + \frac{1}{3}\delta^{ij} Y_{ij} = 0.$$

Tracelessness of Y_{ab} gives

$$-Y_{00} + \delta^{ij} Y_{ij} = 0,$$

so both Y_{00} and the spatial trace $\delta^{ij} Y_{ij}$ vanish. The remaining condition is therefore

$$\hat{n}^i \hat{n}^j Y_{ij}^{\mathrm{TF}} = 0 \qquad \text{for all } \hat{n} \in S^2,$$

where Y_{ij}^{TF} is the traceless spatial part. The $\ell = 2$ spherical harmonics are linearly independent, so this implies $Y_{ij}^{\mathrm{TF}} = 0$. Hence $Y_{ab} = 0$ and therefore $X_{ab} = \phi g_{ab}$. □

Remark 6.13 (Claim boundary in the null bridge)**.** *The c.12-local bridge ends with an explicit theorem-level object: the positive self-adjoint null-translation generator P_Ω on its Stone domain, together with its Borchers affine covariance relation and the half-line modular identities above. What remains unproved here is exactly what the compact note marks as downstream: the density upgrade from the weak endpoint derivative to a local operator-valued distribution, the relativistic identification of that density with continuum T_{kk}, and any bounded-interval modular-Hamiltonian formula beyond the separate interval-preserving projective branch isolated in Theorem 6.11. Positivity and unitary implementation of null translations belong to the established bridge output, while the density upgrade and null-stress identification remain additional premises.*

6.3 Generalized entropy and the Einstein equation

From this point through Corollary 6.18 we specialize to the physical $d = 4$ scaling branch.

For a reference state ω and a small cap C,

$$\delta S_C = \delta\langle K_C \rangle.$$

By Theorem 6.1,

$$\delta S_C = 2\pi\, \delta\langle B_C \rangle.$$

By Proposition 5.10,

$$S_{\mathrm{gen}}(C) = \mathrm{Tr}(\rho L_C) + S_{\mathrm{bulk}}(C).$$

Together with Definition 5.11, this identifies the area term as the coarse-grained form of the edge entropy density rather than an independent postulate.

Definition 6.14 (Admissible equilibrium variation)**.** *Throughout this subsection, δ denotes a first-order state variation about a reference state ω, within the cap-label-preserving MaxEnt class, at fixed cap C, fixed cap size/volume, and fixed conserved charges. Shape or null-cut deformations are denoted separately by δ_{shape} or ∂_λ.*

Assumption 6.15 (Fixed-cap generalized-entropy stationarity)**.** *For admissible variations preserving cap labels, cap size/volume, and relevant conserved charges, the realized reference state is stationary for generalized entropy:*

$$\delta S_{\mathrm{gen}}(C) = 0.$$

This is the Jacobson-type fixed-cap stationarity premise used in the compact note. MaxEnt alone does not derive it here.

Lemma 6.16 (d=4 small-ball modular bridge)**.** *Assume the hypotheses of Theorem 6.11 and a locally Lorentzian scaling regime in which $\delta\langle T_{00}\rangle$ is approximately constant across a small radius-ℓ ball/diamond centered at the cap point. Assume moreover that the transported D4 carried remainder contributes only*

$$\delta\langle E^{(\eta)}_{C,\ell}\rangle = o(\ell^4)$$

to the cap-local first-variation matrix element along the same scaling family. Then in $d = 4$,

$$\delta S_C = 2\pi\,\delta\langle B_C\rangle = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\,\delta\langle T_{00}\rangle\, d^3x = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4).$$

Proof sketch. Theorem 6.11 supplies the null modular/stress-tensor identification in the scaling branch together with the carried remainder $E^{(\eta)}_{C,\ell}$. By the additional hypothesis of the lemma, that remainder contributes only $o(\ell^4)$ to the first variation relevant for the small-ball comparison. In the locally Lorentzian $d = 4$ limit, the cap-preserving conformal generator is the standard ball modular kernel $(\ell^2 - r^2)/(2\ell)$. Integrating that kernel over the three-ball gives

$$\int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, d^3x = \frac{4\pi\ell^4}{15},$$

and multiplying by the overall 2π factor from the first law yields the stated coefficient, with the geometric-gradient term and the carried D4 remainder left explicit at orders $O(\ell^5\partial T)$ and $o(\ell^4)$. □

Theorem 6.17 (Conditional Jacobson-type Einstein relation from entanglement equilibrium)**.** *Assume Axioms 3.1–3.4, Assumption 3.6, Assumption 6.15, and the hypotheses of Theorem 6.11 and Lemma 6.16. Then, for every sufficiently small cap C and every admissible first variation δ about a reference state ω in the locally Lorentzian $d = 4$ scaling regime,*

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame.

Proof sketch. Assumption 6.15 gives the fixed-cap stationarity condition

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Lemma 6.16 identifies the bulk modular contribution in the $d = 4$ small-ball regime as

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4).$$

For a fixed-volume small ball in $d = 4$, the standard area variation is

$$\delta A\big|_{V,\Lambda_{\mathrm{ref}}} = -\frac{4\pi\ell^4}{15}\,\delta(G_{00} + \Lambda_{\mathrm{ref}} g_{00})\,.$$

Since $S_{\text{gen}} = S_{\text{bulk}} + A/(4G)$ in the coarse-grained scaling dictionary, stationarity gives

$$0 = \delta S_{\text{bulk}} + \frac{\delta A}{4G} = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4) - \frac{\pi\ell^4}{15G}\,\delta(G_{00} + \Lambda_{\text{ref}} g_{00})\,,$$

and dividing by ℓ^4 before taking the small-ball limit removes the $O(\ell^5\partial T) + o(\ell^4)$ contribution, leaving the stated rest-frame relation. □

Corollary 6.18 (Tensor Upgrade in the Scaling Regime)**.** *If the first-variation relation of Theorem 6.17 holds for all local directions and reference states in the scaling regime, then*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\,\langle T_{ab}\rangle$$

modulo the metric-term ambiguity already isolated by Lemma 6.12.

Proof. Overlap consistency across observers through the same point supplies all timelike directions, and Theorem 6.11 together with Lemma 6.12 upgrades the rest-frame relation to the full tensor equation in the Jacobson entanglement-equilibrium line of argument [**?**, **?**]. □

6.4 The cosmological constant

Lemma 6.19 (Vacuum-Energy Blindness)**.** *For any null vector k and any vacuum-energy contribution $T^{\text{vac}}_{ab} = -\rho_{\text{vac}} g_{ab}$,*

$$T^{\text{vac}}_{kk} = 0.$$

Proposition 6.20 (Structural Separation of Λ)**.** *Local null-modular data fix T_{ab} only up to ϕg_{ab}. Therefore Λ is not determined by local overlap consistency and requires a global input.*

Corollary 6.21 (Cosmological Constant from Capacity)**.** *If the global screen capacity is identified with the de Sitter static-patch entropy,*

$$N_{\text{scr}} = S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

then

$$\Lambda = \frac{3\pi}{G N_{\text{scr}}}.$$

Proposition 6.20 and Corollary 6.21 are the intended local/global D6 package: local null-modular reconstruction leaves the Einstein equation determined only up to a metric term, and the global screen-capacity identification fixes the remaining constant. This is a structural aspect of the framework. The vacuum-energy problem is reorganized by separating local null data from the global capacity term.

7 Gauge Reconstruction and Standard Model Structure

7.1 Compact gauge reconstruction

At any fixed UV cutoff, edge-center completion equips collars with finitely many sector labels and fusion rules. The category used for reconstruction is the refinement-stable directed colimit

$$\mathsf{Sect}_\infty := \varinjlim_{r} \mathsf{Sect}_r$$

over regulator refinements r, retaining only transportable sectors and intertwiners that persist under refinement. This is the object on which Tannaka/Doplicher-Roberts reconstruction is applied.

Theorem 7.1 (Conditional compact gauge reconstruction in the bosonic branch)**.** *Assume Axioms 3.1–3.4, the relevant parts of Assumption 3.6, and that the refinement-stable edge-sector category* Sect_∞ *in the EFT regime is a rigid symmetric* C^**-tensor category with a faithful bosonic fiber functor to finite-dimensional Hilbert spaces. Then*

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$$

for a compact group G *uniquely determined up to isomorphism.*

Proof sketch. Each fixed-cutoff category has only finitely many simple blocks, but the refinement-stable colimit Sect_∞ can have infinitely many simple objects while still carrying objectwise finite-dimensional fibers. In the bosonic EFT branch, symmetry and rigidity place the category in the Doplicher-Roberts/Tannakian reconstruction class, and the faithful bosonic fiber functor identifies it with $\mathrm{Rep}(G)$ for a compact group G [**?**, **?**]. □

If a fermionic sign object is present, the correct reconstruction statement is super-Tannakian rather than purely Tannakian. The compact note does not prove that fermionic/chiral extension; it works only in the bosonic internal-gauge branch.

7.2 Minimal admissibility and the Standard Model quotient

Throughout this subsection, χ_{cpl} means the dimension of the minimal coupled carrier supporting the required weak-type and color-type nonabelian charges on a common block. It is not the abstract minimal faithful representation dimension of the final gauge group.

Definition 7.2 (Weak-type and color-type nonabelian roles)**.** *A weak-type nonabelian charge is a pseudoreal nonabelian doublet role. A color-type nonabelian charge is an intrinsically complex nonabelian role, meaning that its nonabelian factor itself acts by an irreducible representation not equivalent to its conjugate. Abelian twisting of a pseudoreal nonabelian irreducible representation does not count as a color-type role.*

The derivation of the Standard Model quotient separates into five steps:

1. existence of a compact reconstructed group;
2. identification of the minimal nonabelian sector content;
3. identification of the minimal coupled carrier;
4. uniqueness of the connected abelian factor once admitted on that carrier;
5. determination of product gauge structure up to finite quotient, then of the global finite quotient from the realized matter spectrum.

Lemma 7.3 (Minimal nonabelian sector content)**.** *Any admissible low-energy sector must contain both a weak-type pseudoreal nonabelian charge role and an intrinsically complex color-type nonabelian charge role.*

Proof. Light chiral matter requires left-handed multiplets and right-handed singlets to carry inequivalent gauge data. A pseudoreal doublet structure is the minimal way to realize the weak sector without immediately producing vectorlike masses. A genuinely complex nonabelian representation is required to distinguish quarks from antiquarks. A single nonabelian simple factor cannot simultaneously supply both a minimal weak-type role and an intrinsically complex color-type role; abelian twisting does not count toward the color-type requirement. □

Lemma 7.4 (Connected 2D pseudoreal image classification)**.** *Let H be a connected compact group with a faithful irreducible 2-dimensional pseudoreal unitary representation. Then the connected derived subgroup of its image is conjugate to* $\mathrm{SU}(2)$*. Equivalently, the nonabelian factor acting in that role is* $\mathrm{SU}(2)$ *up to finite central quotient.*

Proof sketch. Irreducibility places the image inside $\mathrm{U}(2)$ with scalar commutant. Its connected derived subgroup is therefore a connected compact subgroup of $\mathrm{SU}(2)$. The connected compact subgroups of $\mathrm{SU}(2)$ are either tori or $\mathrm{SU}(2)$ itself. The representation is nonabelian and pseudoreal, so the torus case is excluded. Hence the derived subgroup is conjugate to $\mathrm{SU}(2)$, with only a finite central kernel left invisible in the representation. □

Lemma 7.5 (Connected 3D intrinsically complex image classification)**.** *Let H be a connected compact group with a faithful irreducible intrinsically complex 3-dimensional unitary representation. Then the connected derived subgroup of its image is conjugate to* $\mathrm{SU}(3)$*. Equivalently, the nonabelian factor acting in that role is* $\mathrm{SU}(3)$ *up to finite central quotient.*

Proof sketch. Because the representation is irreducible on $\mathbb{C}^3$, the semisimple part of the Lie algebra of the image acts irreducibly on $\mathbb{C}^3$. A product of two nontrivial simple factors would force a tensor-product decomposition of dimension at least 4, so only one simple factor can occur. Among compact simple Lie algebras, the only ones with nontrivial irreducible representations of dimension at most 3 are $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$. The 3-dimensional $\mathfrak{su}(2)$ representation is real, not intrinsically complex, whereas the fundamental $\mathfrak{su}(3)$ representation is intrinsically complex. Therefore the connected derived subgroup is conjugate to $\mathrm{SU}(3)$, again up to finite central kernel. □

Lemma 7.6 (Minimal coupled carrier under MAR)**.** *Within the connected positive-dimensional Lie admissible class used by MAR, and under the weak-type / intrinsically complex color-type criteria above, the minimal coupled carrier supporting both roles on a common block has the form*

$$V = \mathbb{C}^3 \otimes \mathbb{C}^2, \qquad \chi_{\mathrm{cpl}} = 6.$$

Proof. By Lemma 7.4, the minimal weak-type nonabelian role is the pseudoreal doublet of $\mathrm{SU}(2)$, hence has dimension 2. By Lemma 7.5, the minimal intrinsically complex color-type nonabelian role is the triplet of $\mathrm{SU}(3)$, hence has dimension 3.

By definition of χ_{cpl}, the weak-type and color-type roles must act nontrivially on one common irreducible nonabelian block rather than on disconnected direct-sum summands. Commuting irreducible actions on such a common block therefore require dimension at least the product of the minimal weak and color dimensions:

$$\chi_{\mathrm{cpl}} \geq 2 \cdot 3 = 6.$$

Equality is achieved by the tensor-product carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$. The block-diagonal representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ is faithful of dimension 5, but it is not coupled and hence is not the MAR minimizer. Connected non-semisimple counterexamples of $U(2)$ type are excluded by the explicit color-type criterion rather than being silently ignored. □

Remark 7.7. *The dimension comparisons used in Lemma 7.6 apply to the positive-dimensional connected Lie image carrying the admissible nonabelian charges, equivalently to the identity component of the reconstructed group on that sector. Finite, disconnected, or connected non-semisimple counterexamples exist, but they are outside the admissible class fixed by the explicit weak-type / color-type criteria.*

MAR supplies existence of a connected abelian factor; the next lemma proves only uniqueness and identification on the minimal coupled carrier.

Lemma 7.8 (Uniqueness of the connected abelian factor on the minimal coupled carrier)**.** *Inside* $\mathrm{U}(6)$*, the commutant of* $\mathrm{SU}(3) \times \mathrm{SU}(2)$ *acting on* $\mathbb{C}^3 \otimes \mathbb{C}^2$ *is exactly* $\mathrm{U}(1)$*.*

Proof. The SU(3) action is irreducible on the first tensor factor and trivial on the second. The SU(2) action is irreducible on the second tensor factor and trivial on the first. By Schur's lemma, any operator commuting with both actions is scalar on each irreducible factor. Hence the full commutant is a single U(1). □

Theorem 7.9 (Product Gauge Structure up to Finite Quotient)**.** *Assume Axiom 3.5, vanishing obstruction* $[z] = 0$ *on the realized low-energy branch, the hypotheses of Theorem 7.1, the weak-type / color-type MAR premises above, and that the realized connected gauge image acts faithfully on the minimal coupled carrier. Then the realized connected gauge structure has the form*

$$G_{\mathrm{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

Proof. Theorem 7.1 yields some compact group G. Lemma 7.3 identifies the minimal admissible sector content, Lemmas 7.4 and 7.5 identify the connected nonabelian factors realizing the minimal weak-type and color-type roles, and Lemma 7.6 fixes the minimal coupled carrier. The tensor-product structure of that carrier implies commuting weak and color actions, so the connected semisimple part of the realized image is locally $\mathrm{SU}(3) \times \mathrm{SU}(2)$. Axiom 3.5 supplies the existence of one connected abelian charge factor acting nontrivially on the coupled carrier, and Lemma 7.8 shows that any such connected abelian factor is necessarily a single U(1). A compact connected Lie group with Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is a quotient of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ by a finite central subgroup. Faithfulness rules out an invisible extra torus, so the only remaining ambiguity is the finite central quotient Γ. □

Theorem 7.10 (Hypercharge lattice on the realized matter package)**.** *Assume gauge group* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of chiral matter* (Q, u^c, d^c, L, e^c)*, one Higgs doublet* H*, and Yukawa terms*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly cancellation and Yukawa invariance determine the hypercharge ratios up to an overall $\mathrm{U}(1)_Y$ *normalization:*

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_c Y_Q.$$

Fixing the normalization by $Q = T_3 + Y$ *and* $Q(\nu_L) = 0$ *gives*

$$Y_Q = \frac{1}{2N_c}.$$

For $N_c = 3$ *one recovers the exact Standard Model lattice*

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_e = 1, \quad Y_H = \frac{1}{2}.$$

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \qquad Y_d = -Y_Q + Y_H, \qquad Y_e = -Y_L + Y_H.$$

The mixed anomalies yield

$$N_c Y_Q + Y_L = 0,$$

and

$$2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0.$$

Solving the first anomaly gives

$$Y_L = -N_c Y_Q.$$

Substituting the Yukawa relations and $Y_L = -N_c Y_Q$ into the mixed gravitational anomaly then yields

$$Y_H = N_c Y_Q, \qquad Y_u = -(N_c+1)Y_Q, \qquad Y_d = (N_c-1)Y_Q, \qquad Y_e = 2N_c Y_Q.$$

With these relations in place, the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ anomalies cancel identically. The normalization of Y_Q is fixed by the electric charge operator $Q = T_3 + Y$ together with $Q(\nu_L) = 0$, which implies

$$Y_L = -\frac{1}{2} = -N_c Y_Q, \qquad \text{hence} \qquad Y_Q = \frac{1}{2N_c}.$$

□

Corollary 7.11 (Three Generations on the realized MAR-admissible branch)**.** *Under the hypotheses of Theorem 7.9, the asymptotic-freedom/CP-capability window together with the realized MAR selector gives*

$$N_g = 3.$$

Proof. Intrinsic CKM CP violation requires

$$\frac{(N_g-1)(N_g-2)}{2} > 0,$$

hence $N_g \geq 3$. For one Higgs doublet, the one-loop weak-sector coefficient is

$$b_2 = \frac{11}{3}C_A - \frac{2}{3}\sum_f T(R_f) - \frac{1}{3}\sum_s T(R_s) = \frac{22}{3} - \frac{N_g(N_c+1)}{3} - \frac{1}{6} = \frac{43}{6} - \frac{N_g(N_c+1)}{3}.$$

Asymptotic freedom therefore implies

$$N_g(N_c+1) < \frac{43}{2}.$$

Since the color-type requirement forces $N_c \geq 3$, this gives $N_g \leq 5$. The realized MAR selector then chooses $N_g = 3$. □

Corollary 7.12 (Three Colors on the realized MAR-admissible branch)**.** *Under the hypotheses of Theorem 7.9, with $N_g = 3$ from Corollary 7.11, Witten's global* SU(2) *anomaly together with the realized MAR selector gives*

$$N_c = 3.$$

Proof. Witten's global SU(2) anomaly requires the total number of left-handed weak doublets to be even:

$$N_g(N_c+1) \equiv 0 \pmod 2.$$

With $N_g = 3$, this implies N_c is odd. The case $N_c = 1$ is inadmissible because it removes the intrinsically complex color sector demanded by Lemma 7.3. Minimal admissibility therefore selects $N_c = 3$ [**?**]. □

Proposition 7.13 (Global quotient from the realized matter spectrum)**.** *If the realized matter spectrum carries the hypercharges of Theorem 7.10 with $N_c = 3$, then the subgroup of* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *acting trivially on all realized states is exactly* $\mathbb{Z}_6$*.*

Proof. Write $q := 6Y \in \mathbb{Z}$ and $\eta := e^{i\pi/3}$, so the U(1) factor acts by η^q on charge-q multiplets. The element

$$g_0 := (\omega_3, -1, \eta)$$

acts trivially on the realized matter and Higgs multiplets:

$$Q: \ \omega_3(-1)\eta = 1, \qquad u^c: \ \omega_3^{-1}\eta^{-4} = 1, \qquad d^c: \ \omega_3^{-1}\eta^2 = 1,$$

$$L: \ (-1)\eta^{-3} = 1, \qquad e^c: \ \eta^6 = 1, \qquad H: \ (-1)\eta^3 = 1.$$

Its powers therefore generate a subgroup of order six acting trivially on all realized states. Conversely, any central element acting trivially on e^c must have U(1) part η^n, and triviality on Q then fixes the SU(3) and SU(2) center factors uniquely. Hence every trivial central element is a power of g_0, so the kernel is exactly $\mathbb{Z}_6$. □

Corollary 7.14 (Standard Model Gauge Group)**.** *Under Theorem 7.9, Theorem 7.10, Corollaries 7.11 and 7.12, and Proposition 7.13,*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Theorem 7.9 yields the product structure up to finite quotient, and Proposition 7.13 fixes that quotient to $\mathbb{Z}_6$ once the realized hypercharge lattice and $N_c = 3$ are in place. □

Corollary 7.15 (Product-Group Consequences)**.** *The adjoint representation of the full connected gauge group contains only*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

Equivalently, the adjoint representation of the derived gauge group contains only

$$(8, 1, 0) \oplus (1, 3, 0).$$

There are no mixed $(3, 2, \pm 5/6)$ *gauge bosons. Hence gauge-mediated proton decay is absent. The unbroken gauge symmetries forbid elementary gauge-boson mass terms: the photon remains massless, while gluons remain massless gauge fields of the unbroken color symmetry.*

Remark 7.16. *The same product-group structure removes the usual grand-unification monopole sector. In the present framework, unification of couplings does not proceed by embedding the gauge group into a simple group that is later broken. It proceeds through a shared geometric origin of the edge heat-kernel parameters. Absence of gauge-mediated proton decay and unification therefore coexist without tension. Any confinement statement is a separate infrared dynamical issue rather than a corollary of the product-group adjoint argument alone.*

8 Relation to Holography and Existing UV Frameworks

OPH belongs to the holographic family of ideas, but its primitive data are different from those used in asymptotic-boundary constructions. The distinction matters because several of the central claims of the present paper, including the treatment of Λ, the handling of factorization, and the emergence of gauge structure, depend on it.

8.1 Static-patch screen versus asymptotic boundary

In asymptotic-boundary holography one starts from a dual theory at infinity and reconstructs the bulk inward. In OPH one starts from a finite-capacity screen equipped with a net of local patch algebras and reconstructs the effective bulk from overlap consistency. The primitive object is therefore a family of observer patches with nontrivial overlaps, not a single global boundary theory.

This shift has two technical consequences. First, subsystem factorization is treated from the beginning as a gluing problem with centers and sector labels. Second, the physical role of the horizon is local and operational: every observer has direct access only to a patch algebra, and global law is whatever survives reconciliation on overlaps.

8.2 Positive Λ and finite capacity

The null-modular reconstruction of Section 6 determines the stress tensor only up to a metric term. This means that local null data do not fix Λ. The cosmological constant enters only when one supplies the global screen capacity

$$N_{\mathrm{scr}} = \log \dim \mathcal{H}_{\mathrm{tot}}.$$

The resulting relation

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}$$

fits naturally with a de Sitter static patch of finite entropy via $N_{\mathrm{scr}} = A_{\mathrm{dS}}/(4G)$. This is the same D6 local/global split in prose: null-modular reconstruction says what local data cannot fix, and screen capacity supplies the missing global datum. OPH is therefore formulated in a de Sitter-first language. The positive cosmological constant is tied to finite capacity rather than to a deformation of a negative-Λ starting point.

8.3 Factorization, gauge structure, and the string sector

The factorization problem of gauge theory and gravity is often treated as a technical nuisance. In OPH it is part of the architecture. Edge-center completion turns the collar center into a first-class object, and the entropy split

$$S_{\mathrm{gen}} = \langle L_C \rangle + S_{\mathrm{bulk}}$$

is then a structural consequence rather than an added prescription.

In the bosonic EFT branch, the same collar data produce a symmetric rigid transportable sector category. Compact gauge reconstruction then follows conditionally from that category, and the Standard Model quotient is selected by minimal admissibility. This differs sharply from approaches in which the gauge group is specified in advance.

The string-theory relation is similarly reversed. The edge partition function

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the closed partition function obtained after gluing the open-edge weights $p_R(t) \propto d_R e^{-tC_2(R)}$. It has the heat-kernel form familiar from two-dimensional Yang-Mills theory. If a large-N_{edge} regime exists, with N_{edge} distinct from the physical color rank $N_c = 3$, its free energy admits a genus expansion. Within OPH this is read as a conditional effective reorganization of edge degrees of freedom, not as a derived critical-superstring completion. Any further lift to critical superstring structure would additionally require worldsheet supersymmetry, critical dimension, modular invariance, and GSO projection beyond the OPH axiom chain.

Feature	Asymptotic-boundary holography	OPH
Primitive data	Boundary theory at infinity	Finite-capacity screen with observer-patch net
Factorization across cuts	Subtle and often indirect	Explicit edge-center completion with central labels
Status of Λ	External background datum	Fixed globally by N_{scr} after local null reconstruction
Gauge group	Usually specified as part of the model	Reconstructed from edge sectors, then filtered by admissibility
String sector	Fundamental dual description	Effective large-N_{edge} reorganization of edge dynamics

9 Predictive Status and Falsifiability

This section is the authoritative falsifiability ledger for the three-phase program. In the compact note, the recovered core is

$$(D1\text{–}D5) \cup (D7\text{–}D9),$$

namely the relativity chain together with the realized Standard Model structural chain. D6 is a separate input-dependent capacity corollary, D10 is an implementation-level supplement-backed calibration branch, and D12 collects deferred continuations.

9.1 Prediction audit by claim tier

Tier	Representative outputs	What would actually falsify the tier
Phase I recovered core (Theorem 3.10; D1–D5, D7–D9)	Confluence of overlap repair, conditional Lorentz kinematics, the conditional Jacobson-type Einstein branch in the stated scaling regime, compact gauge reconstruction in the bosonic branch, the realized Standard Model quotient chain, the exact hypercharge lattice on the realized matter package, the realized counting chain $N_g = 3$ then $N_c = 3$, and the product-group consequence of no gauge-mediated proton decay	A mathematical failure in the derivation chain, or data that require a different realized gauge quotient, different realized hypercharges, a different color or generation count on the admissible branch, or gauge-mediated proton decay
Phase II input-dependent corollary (D6)	$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}$ once the screen-capacity identification is made, together with the separate order-of-magnitude neutrino estimate tied to the same capacity input	Failure of the screen-capacity identification or of the specific global capacity relation. Such a failure would not by itself erase the recovered core
Phase II current implementation: calibration sector (D10)	Pixel-closure gauge-coupling consistency, $\alpha_i(m_Z)$, $\alpha_{\mathrm{em}}^{-1}(m_Z)$, $\sin^2\theta_W(m_Z)$, and the present supplement-backed W/Z carrier diagnostics	Failure of the supplement-backed closure of the present implementation once P and the printed running/matching conventions are imposed

Tier	Representative outputs	What would actually falsify the tier
Phase III deferred continuations (D12 and beyond)	Flavor ansätze, charged-lepton/Koide fits, texture branches, neutrino mass/mixing constructions beyond the D6 estimate, dark-sector response laws, baryogenesis estimates, proton-spin bookkeeping, proton-lifetime estimates beyond the gauge-channel exclusion, black-hole spectroscopy templates, conditional string/worldsheet reorganizations, conjectural critical-superstring extensions, and other downstream phenomenology	Retraction of the corresponding continuation only. These branches are not part of the compact paper's recovered core and their failure is not, by itself, a falsification of relativity-plus-Standard-Model recovery

This tier split is the sync rule for mirrored status surfaces: Phase I is the recovered-core falsifiability standard, Phase II is visible but non-core, and Phase III is explicitly continuation-level.

Mirrored interpretive surfaces may discuss a separate OPH-internal state-and-law habitat theorem for the strange-loop wrapper. That result, when stated carefully, still sits outside Phase I and does not change this falsifiability ledger: the missing ingredients remain the actual OPH closure map, an invariant admissible observer-supporting sector for that map, and any uniqueness or stability estimate.

9.2 Recovered-core discriminators

Once the claim boundary is made explicit, the sharpest discriminators are the ones that do not depend on auxiliary flavor, dark-sector, baryogenesis, or string ansätze.

9.2.1 Realized Standard Model gauge structure

On the realized MAR-admissible branch, the compact note fixes

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

with the exact one-generation hypercharge lattice. Here the realized counting chain is sequential: the CP/UV admissibility window and minimal admissibility yield $N_g = 3$, and Witten's SU(2) anomaly parity with that realized generation count yields $N_c = 3$.

$$Y_Q = \frac{1}{6}, \qquad Y_L = -\frac{1}{2}, \qquad Y_u = -\frac{2}{3}, \qquad Y_d = \frac{1}{3}, \qquad Y_e = 1, \qquad Y_H = \frac{1}{2}.$$

Evidence that the realized low-energy gauge structure or realized hypercharges differ from these values would directly contradict the recovered core.

9.2.2 Product-group corollary

Because the realized gauge structure is a product group up to the finite central quotient already fixed above, there are no mixed $(3, 2, \pm 5/6)$ gauge bosons. Hence

$$\tau_p^{(\text{gauge})} = \infty$$

for gauge-mediated proton decay. Observation of proton decay specifically attributable to gauge-boson exchange would therefore falsify the realized product-group branch.

9.2.3 Relativity branch

The relativity claim is conditional but still sharp about its scope: if the stated geometric-modular, null-bridge, and fixed-cap-stationarity premises are realized in the intended scaling regime, then the compact note predicts Lorentz kinematics on the screen and a Jacobson-type Einstein branch. What is being tested here is the existence of that scaling regime and its premises, not a fixed-cutoff matrix-algebra identity.

9.3 What does *not* count as a contradiction of the recovered core

The following do *not* falsify Theorem 3.10 if they fail in their present form:

1. the uniform $\mathbb{Z}_6$ center-label ensemble and the associated $\varepsilon = 1/6$ flavor ansatz;
2. Koide/charged-lepton constructions, texture exponents, or capacity-level neutrino estimates;
3. modular-anomaly dark-sector response ansätze, including MOND/RAR-style scaling claims;
4. baryogenesis estimates based on extra suppression-counting assumptions; a baryogenesis theorem would still need a concrete out-of-equilibrium mechanism, derived CP-odd source terms, justified defect/sphaleron counting, washout control, and a freeze-out computation of the final asymmetry;
5. proton-spin ansätze or proton-lifetime estimates beyond the structural exclusion of gauge-mediated decay;
6. discrete-horizon spectroscopy templates;
7. large-N_{edge} string/worldsheet reorganizations.

These are all post-Phase-I, non-core branches. Some are Phase-II input-dependent estimates, while the rest are Phase-III continuations. None is part of the compact note's recovered relativity-plus-Standard-Model core.

9.4 What the compact paper claims after scope-cleaning

After the present scope clean, the compact paper claims exactly this: observer-overlap consistency, together with the stated scaling-limit and categorical premises, recovers a conditional relativity branch and the realized Standard Model structural branch. The Phase-II capacity relation and the current Phase-II D10 calibration sector remain visible but secondary. Everything in Phase III is explicitly non-core, including the modular-anomaly dark-sector branch, which remains future work rather than a theorem-level output.

10 Common Objections and Clarifications

10.1 Is the use of P circular?

The answer depends on which sector is under discussion. In the current public formulation the pixel area P calibrates a separate gauge-coupling sector. Quantities algebraically entangled with that calibration are therefore consistency checks for the implementation rather than part of the recovered-core claim set.

Operationally, changing P moves the calibration family $(\alpha_U, \alpha_i(m_Z), \sin^2\theta_W(m_Z))$. It does not alter the parameter-free structural outputs such as the gauge quotient or hypercharge lattice. Downstream matter-sector continuations, including Koide-style relations, remain outside the compact paper's recovered-core theorem package.

10.2 Does a fixed UV cell size break Lorentz invariance?

The UV regulator is placed on the screen algebra, not on an emergent bulk spacetime lattice. Lorentz kinematics in the present framework arise from the continuum modular action on caps, summarized in Corollary 6.3, rather than from microscopic invariance of a bulk discretization.

This is conceptually similar to other continuum limits in statistical mechanics and lattice field theory. A regulator may break a symmetry at finite cutoff while the infrared fixed point restores it exactly. In OPH the restoration is stronger than a generic RG expectation because the modular-flow theorem identifies the symmetry group itself:

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1).$$

Residual Lorentz violation would therefore have to arise from a failure of the regulator premises or from a failure of the modular-flow theorem, not merely from the existence of a UV cell size. The remaining UV-side burden on the Lorentz branch is therefore the explicit branch-selection and scaling-limit package **T4**+**T8**, not the mere existence of a finite cell size.

10.3 Why use type-I algebras if continuum QFT uses type III?

The type-I assumption is a regulator statement. At finite cutoff the local patch algebras are finite-dimensional matrix algebras, which makes entropy and recovery theory well defined. The continuum limit is expected to recover the familiar type-III behavior of local QFT algebras. There is no contradiction here. The two statements refer to different scales.

This is no more problematic than using a lattice regulator for Yang-Mills theory and then taking a continuum limit in which the operator algebra changes type. The mathematical burden is the existence of a controlled scaling regime. In the present manuscript that burden is isolated in Assumption 3.6. It is not hidden inside the main theorems.

10.4 What is the current UV-completion claim?

At fixed cutoff, Axiom 3.3 already yields a quasi-local finite-range interacting branch, and Proposition 4.6 makes the physical normal form unique only after quotienting by boundary redundancy. The present suite therefore does not claim a unique microscopic UV completion. What is fixed is a unique physical branch modulo implementation hiding. On the ordinary or central-defect branch, the remaining fixed-cutoff topological obstruction is the genuinely noncentral 2-group defect singled out above. Beyond fixed cutoff, the continuum modular/geometric lift remains frozen behind **T4** and **T8**, and the transportable bosonic gauge lift still carries **T7** and **T10**–**T12** when invoked.

10.5 Why is charge quantization possible without a simple GUT?

The framework uses the global quotient and anomaly structure rather than a simple-group embedding. Once the realized matter content is fixed, the subgroup acting trivially on all states is exactly $\mathbb{Z}_6$, and the anomaly equations force the sixth-integer hypercharge lattice. Integer electric charge for color singlets is then a property of the realized quotient structure. A simple unified gauge group is not required.

11 Discussion

The framework is strongest where the outputs are discrete or structurally rigid:

1. the Lorentz branch inside the assumed geometric modular phase;
2. the conditional scaling-limit Einstein branch under fixed-cap stationarity and the null-stress premise;
3. the Standard Model gauge quotient chain;
4. exact hypercharges on the realized one-generation matter package with one Higgs doublet;
5. the realized counting chain $N_g = 3$ then $N_c = 3$ on the realized MAR-admissible branch;
6. the compact two-input economy of the present quantitative implementation, kept explicit as a secondary layer rather than as part of Phase I.

At the same time, the UV status is sharper than a generic "open gap" label. The local MaxEnt branch already gives a finite-range interacting fixed-cutoff dynamics, and Proposition 4.6 gives a unique physical branch modulo implementation hiding. On the ordinary or central-defect branch, the fixed-cutoff topological sector is reduced to the named noncentral 2-group obstruction. What remains open is the continuum BW/geometric lift of **T4**+**T8** and the fuller transportable compact-gauge lift when **T7** and **T10**–**T12** are invoked.

The next strongest non-core sector is the supplement-backed D10 calibration branch. On the present local implementation, the current D10 carrier closes its own exact W/Z chart, but exact PDG W/Z does not yet honestly follow on that carrier; the remaining quantitative burden should therefore be read as a repair branch beyond the present carrier rather than as a hidden extra decimal place or an unnamed selector on the existing point. The separate D6 cosmological-capacity relation remains visible as an input-dependent corollary rather than as part of the recovered core. The charged-lepton and Koide sectors remain phenomenological continuations rather than core theorem outputs; the strongest current local charged-lepton result is an explicit same-carrier obstruction together with a two-scalar support-extension shell, not a closed emitted spectrum. On the flavor side, the only structural output already fixed before those continuations is the realized counting chain $N_g = 3$ then $N_c = 3$.

The weakest part is the light-quark and hadron branch. Current local tier audits indicate that a nonzero light-quark pure-B selector is not available at recovered-core tier in the present corpus, so the honest light-quark split frontier remains a D12 continuation branch rather than a hidden core theorem. The hadron lane is weaker still: its present frontier is nonperturbative runtime completion plus stable-channel convergence rather than a closed theorem-level mass surface.

The framework should therefore be evaluated in stages. It already contains several results that are normally postulated independently in other approaches: kinematics, dynamics, gauge structure, charge lattice, matter multiplicities, symmetry-protected massless sectors, and a limited downstream quantitative surface in supplement-backed or continuation-level form. The strongest current local descendant beyond D10 is a compact Higgs/top forward seed on the same gauge core, but that branch also remains secondary to the recovered core. Whether one regards this as evidence for fundamentality depends on whether these simultaneous derivations survive further scrutiny.

Topics intentionally excluded from the present formulation include simulation narratives, observer-continuation mechanisms, and claims of final completeness for all low-energy observables.[2]

[2]Questions outside this theorem package do not alter the specific claim set established here. Even a separate

12 Conclusion

Observer-Patch Holography starts from overlap consistency of observer patches and yields, within one framework, several effective structures usually treated separately in modern fundamental physics. The compact note supports a unique schedule-independent normal form for overlap repair and hence a unique physical UV branch modulo boundary redundancy and implementation hiding, a conditional Lorentz branch in the assumed geometric modular phase, a conditional Jacobson-type Einstein branch under fixed-cap stationarity and the null-stress premise, the Standard Model gauge quotient with exact hypercharges on the realized matter package under the MAR admissibility package in the bosonic internal-gauge branch, the realized counting chain $N_g = 3$ then $N_c = 3$ on that branch, a separate supplement-backed gauge-calibration sector that remains secondary to the recovered core, and a separate input-dependent cosmological-capacity corollary. It also records several phenomenological continuations whose present status is weaker and is labeled as such in the text.

Accordingly, the framework functions as a candidate underlying theory from which general relativity and the Standard Model may arise as effective descriptions once the stated scaling-limit and categorical premises are realized. The next technical tasks are sharper quantitative control, derivation or reduction of the noncentral topological obstruction and the transportability-side premises, a fuller fermionic gauge branch, and explicit microscopic realizations whose continuum lift lands in the required modular/geometric phase.

interpretive habitat theorem for OPH state-and-law data would not enlarge the compact paper's Phase-I scope unless the actual closure map, invariant admissible sector, and stability hypotheses were also proved at the same tier.